1. DDU PROFILE

Dharmsinh DesaiUniversity

ISO 9001:2008

Nadiad - 387 001, Gujarat, India.

University (formerly known as DDIT), was established in June 1968 by Late Shri Dharmsinh Desai (then Member of Parliament), with a view to provide quality education in Engineering and Technology to the youth of Gujarat and was affiliated to Gujarat University.

Having modesty started with Diploma and Degree courses in Chemical Engineering. It has by now grown as top most Institute of Gujarat offering Graduate and Post Graduate level courses in various disciplines, and ultimately has become a State University in the year 2005. The institute gained an autonomous status in 1999 with approval from UGC and Gujarat Government and become a Deemed University accredited by N.B.A. in the year 2000.This University offers Graduate and Post Graduate level coursesin various disciplines and Master of Computer Applications (MCA) is being run under Faculty of Management and Information Science since 1987.

The University has also received ISO-9001:2008 Certification from ISOQAR and have also acquired NAAC accreditation. D.D.University believes not only in providing quality education, but also provides an overall development of its students in the form of being disciplined, sociality accountable to the society and responsible citizen of our county.

D.D. University is being guided by the vision laid down by its founding Chairman Late Shri Dharmsinh Desai, the present Chairman Dr. Narendra Desai and under a strong and dynamic leadership of Vice-chancellor Dr. H.M.Desai.

Vision:
"Become a multi-disciplined & „learner oriented" university to closely associate with and be responsive to the industry, to create supportive and caring environment for staff and students and to engage in R & D activities in areas of national priority."

Mission:
"Undertake programs and projects for development of human resources, both through formal and non-formal delivery systems, in areas of professional pursuits in all walks of human endeavors, with accent on relevance, value addition, societal needs and futuristic pilot projects."

2. COMPANY PROFILE

ROIMA

Company name: Roima Intelligence India private limited

Company address: Satyamev Elite, 608 - 615, Sardar Patel Ring Rd, South Bopal, Bopal, Ahmedabad, Gujarat 380058

Company Email: neha.vindhani@roimaint.com

Company Number: +91 7069692727

About us: Roima Intelligence is a global business-solutions provider specializing in digital transformation for manufacturing, supply chain, logistics, and product lifecycle management.The company delivers a broad portfolio of software and services including ERP, MES/MOM, WMS, PLM, logistics platforms, machine vision, and more to help organizations streamline operations, integrate systems, and manage supply chains effectively.With more than 500 customers worldwide and over 400 professionals working across several countries, Roima brings deep industry expertise and a track record of successful implementations.Roima aims to be a reliable, long-term partner not just a vendor helping clients modernize their infrastructure, improve quality, and operate with greater efficiency and resilience.

Mission: Roima's mission is to deliver powerful, integrated software that helps businesses improve productivity, quality, and efficiency. They focus on creating measurable, long-lasting results through supply-chain, manufacturing, and logistics solutions. Roima works as a long-term partner, guiding clients through digital transformation and ensuring their systems run smoothly, sustainably, and profitably.

Vision: Roima's vision is to lead global digital transformation by helping businesses build smart factories and intelligent supply chains. They aim to make operations fully connected, automated, and data-driven so companies can work faster, smarter, and more efficiently. Roima wants to expand internationally and continue innovating solutions that strengthen competitiveness and prepare organizations for the future.

3. PROJECT PROFILE

Project Title	HR Management System
Company Name	Roima Intelligence India private limited
Tools	Postman, Visual Studio, Vs Code, SQL Server Management Studio
Duration	19th January, 2026 to 21th March, 2026
Team Size	1
Team Member	Kartik Miteshbhai Patel
Internal Guide	Associate Professor Minal V. Shah
External Guide	Yash Shah

4. TOOLS & TECHNOLOGY

Tools :-

Sr. No	Tools	Purpose
1	Visual Studio	Writing Code - Backend
2	Visual Studio Code	Writing Code – Frontend
3	SQL Server Manager Studio	Database
4	Postman	API Testing

Technology :-

Sr. No	Technology	Purpose
1	Asp.net – C#	Backend Technology
2	React.js – TypeScript	Frontend Technology
3	SQL Server Manager Studio	Database

5. PROJECT PLANNING

5.1 Project Definition :

The HR Management System (HRMS) is a comprehensive, web-based enterprise application developed to digitize, streamline, and automate the core human resource functions of an organization. In today's fast-paced corporate environment, managing HR activities through manual methods such as spreadsheets, paper forms, and email chains leads to significant inefficiencies, errors, and delays. The HRMS addresses these challenges by providing a unified digital platform where all stakeholders Employees, Managers, and HR Administrators can interact with HR processes in a structured, transparent, and efficient manner.

The system is designed around six core functional areas: Travel Management, Expense Management, Social Achievements and Celebrations, Game Slot Scheduling, Organization Chart Visualization, and Job Listing with Referral Management. Each of these modules is tightly integrated with a role-based access control system, ensuring that every user only sees and interacts with the features and data relevant to their role. This not only improves security but also reduces confusion and improves the overall user experience.

From an HR perspective, the system empowers HR Administrators to take full control of organizational activities. They can create and assign travel plans to employees, set expense limits, verify and approve submitted expense claims, configure game slot timings, manage job openings, review referrals, and monitor the overall organizational hierarchy. All of these tasks, which previously required significant manual coordination across multiple tools, can now be accomplished from a single, centralized dashboard.

From an employee perspective, the system provides a self-service portal where employees can view their assigned travel plans, upload necessary travel documents, submit expense claims with proof attachments, participate in social achievement activities, book game slots during leisure time, refer friends and former colleagues for job openings, and stay updated with in-app and email notifications. This level of self-service reduces dependency on HR staff for routine queries and empowers employees to manage their own information efficiently.

From a managerial perspective, Managers have a read-only view of their team's activities. They can monitor travel assignments, review expense submissions, view the organizational chart, and keep track of team members' engagement. This gives managers the visibility they need to support their teams without interfering in the operational decisions that HR controls.

The system is built with a modern, scalable technology stack ASP.NET Core Web API on the backend, React.js with TypeScript on the frontend, and SQL Server as the database. JWT-based authentication ensures secure login, while Entity Framework Core handles all database interactions efficiently. The application is designed to be responsive, meaning it works seamlessly across desktop, tablet, and mobile browsers.

Example:

To better understand how the system works in practice, consider the following real-world scenario. An employee named Rahul is assigned to travel to Mumbai for a client meeting. The HR Administrator logs into the system, creates a travel plan with details such as start date, end date, destination, and maximum expense limit, and assigns it to Rahul. Rahul receives an in-app notification and an email alert informing him of the travel assignment.

During and after the trip, Rahul uploads his travel documents such as flight tickets and hotel invoices directly through the system. After the trip is completed, Rahul submits his expense claims one by one, each linked to the specific travel plan, along with proof documents such as scanned receipts. The HR Administrator reviews each submitted expense, checks the proof documents, and either approves or rejects them with remarks.

At the same time, Rahul participates in the social feed by posting about a project achievement his team just completed. His colleagues like and comment on the post. The system also auto-generates a birthday post for a colleague, which is visible to everyone in the organization.

When Rahul has some free time, he books a game slot for a table tennis session through the game scheduling module. Since fair allocation is enforced, the system ensures that no single employee monopolizes the slots. Later, Rahul notices a job opening posted by HR that perfectly fits a former colleague's skills. He uses the referral module to submit a referral with the candidate's name, email, resume, and a personal note.

Throughout all of this, Rahul can track the status of his expense claims in real time, view the organizational chart to understand reporting structures, and receive timely notifications all from the same platform. This integrated experience is the core value proposition of the HRMS.

5.2 Preliminary Investigation :

Before the development of the HR Management System began, a detailed preliminary investigation was conducted to understand the existing challenges faced by the organization in managing its human resource operations. This investigation involved analyzing current workflows, identifying pain points, and determining the need for a digital solution.

- **Manual Paperwork and Process Delays:** Traditional HR processes rely heavily on physical paperwork and manual approvals. Travel requests are communicated via email, expense forms are filled out on paper, and approvals require physical signatures. This results in significant delays, especially when approvers are unavailable, and increases the risk of documents being lost or misplaced. A digital system with automated notification and approval workflows directly solves this problem by reducing turnaround times from days to hours.
- **Lack of a Centralized Employee Platform:** Employees currently have no single platform to manage all their HR-related activities. Travel information is communicated via email, expense forms are submitted separately, and engagement activities are organized through WhatsApp groups or notice boards. This fragmentation makes it difficult for employees to track their own information and increases frustration.
- **Difficulty in Managing HR Data Across Scattered Systems:** HR teams often manage employee records, travel plans, expense data, and job postings across multiple disconnected tools such as Excel sheets, email inboxes, and standalone applications. This makes it extremely difficult to get a consolidated view of organizational activities. Data inconsistencies, duplication of effort, and inability to generate reports are common problems. The HRMS centralizes all data in a single SQL Server database with structured modules, making data management significantly easier and more accurate.
- **Need for a Centralized Digital Platform:** The investigation confirmed that the organization urgently needed a centralized digital platform that could automate and streamline all core HR operations. The platform needed to support multiple user roles, enforce business rules automatically, provide real-time status tracking, and integrate communication through notifications. The HRMS was designed specifically to address this need, providing a structured and scalable solution.
- **Automated Notifications for Timely Communication:** One of the key findings of the preliminary investigation was that employees frequently missed important updates about their travel assignments, expense approvals, and referral statuses because communication was manual and inconsistent. The HRMS addresses this by implementing automated email and in-app notifications triggered at every key event such as when a travel plan is assigned, an expense is approved or rejected, or a referral status is updated ensuring that all stakeholders are always informed in real time.
- **Need for Role-Based Access and Data Security:** The investigation also revealed that the existing systems did not enforce proper access control. Employees could sometimes access sensitive HR data that was not meant for them, and there was no audit trail of who accessed or modified what information. The HRMS addresses this through JWT-based authentication and a strict role-based access control system, ensuring that each user can only access the features and data relevant to their role, while all activities are logged for transparency and compliance.

6. SYSTEM ANALYSIS

6.1 Feasibility Study :

6.1.1 Technical Feasibility :

The technical feasibility study evaluated whether the proposed HRMS could be built using available technologies, tools, and expertise within the given timeframe and constraints. The conclusion is that the system is fully technically feasible, and the following points elaborate on why.

- **Modern Microsoft Technology Stack:** The proposed HRMS is built on a modern, industry-proven Microsoft technology stack. The backend uses ASP.NET Core Web API, which is a high-performance, cross-platform framework for building RESTful APIs. It supports dependency injection out of the box, making the codebase modular and testable. The framework also provides built-in middleware for authentication, authorization, error handling, and logging, significantly reducing development effort for common tasks.

- **Scalable and Type-Safe Frontend:** The frontend is developed using React.js with TypeScript. React's component-based architecture enables the development of reusable UI components, while TypeScript adds static type checking that catches errors at compile time rather than runtime. This combination results in a frontend that is not only visually responsive but also maintainable and scalable as new features are added over time. The use of React's virtual DOM ensures fast and efficient rendering even when large amounts of data are displayed.

- **Secure Authentication and Authorization:** Security is a critical requirement for any enterprise HR system. The HRMS implements JWT (JSON Web Token) based authentication, where each user receives a signed token upon login. This token is validated on every API request, ensuring that only authenticated users can access the system. Role-based authorization is layered on top, so even authenticated users can only access endpoints and data appropriate to their role Employee, Manager, or HR Administrator. This dual-layer security model is technically sound and widely adopted in enterprise applications.

- **ORM-Based Database Access with Entity Framework Core:** The HRMS uses Entity Framework Core as its Object-Relational Mapper (ORM), which abstracts away raw SQL queries and allows developers to interact with the SQL Server database using strongly typed C# objects. This approach reduces the risk of SQL injection attacks, speeds up development, and ensures consistent data access patterns across all modules. Migrations are used to manage database schema changes in a controlled and versioned manner.

- **File Upload and Document Management:** Several modules in the HRMS require file uploads expense proofs, travel documents, CV attachments for job referrals, and job description PDFs. The system handles these through multipart form data APIs with strict file type and size validation. Uploaded files are stored in a structured manner with references saved in the database, ensuring efficient retrieval and management. This approach is technically proven and reliable for enterprise document management scenarios.

- **Email Notification Services**: The system integrates an email notification service to automatically send emails to users when key events occur such as travel assignment,

expense approval or rejection, referral submission confirmation, and HR moderation actions on posts. This is implemented using SMTP-based email sending within the ASP.NET Core service layer, which is a mature and well-supported approach for enterprise notification systems.

- **Business Rule Engine in Service Layer:** All critical business rules such as expense submission windows (only within 10 days of travel end date), mandatory proof documents for expenses, fair game slot allocation logic, and referral email routing are implemented in the service layer of the backend. This ensures that business logic is centralized, testable, and independent of the UI layer, making it easier to modify rules in the future without affecting the frontend.

- **Queue-Based Game Slot Scheduling:** The game slot scheduling module uses a queue-based fair allocation algorithm. When multiple employees request the same game slot, the system uses a combination of historical game play data and request order to determine who gets priority. Employees who have played less recently are given higher priority, ensuring fairness. This logic is implemented entirely in the backend service layer and is transparent to the end user.

- **Audit Logging and Status Tracking:** The system maintains a complete audit trail of all critical operations expense status changes, document uploads, referral submissions, and post moderation actions. Each record includes timestamps, the user who performed the action, and the nature of the change. This ensures full transparency and supports compliance requirements.

6.1.2 Operational Feasibility :

The operational feasibility study assessed whether the HRMS would be practically usable by the intended users in the organization's day-to-day operations. The findings confirm that the system is highly operationally feasible.

- **Intuitive and Role-Specific User Interface:** The HRMS is designed with a clean, role-specific interface that presents each user only with the features and information relevant to their role. An employee logs in and immediately sees their travel assignments, expense status, post feed, and game slot availability. An HR Administrator sees a comprehensive dashboard with controls for all HR operations. This targeted design reduces the learning curve significantly and ensures that users can become productive with minimal training.

- **Comprehensive HR Administration Controls:** HR Administrators have full operational control over the entire system. They can create and manage travel plans with multiple travellers, set and enforce expense limits, verify expense claims with supporting documents, configure game slot timings and capacities, create and publish job openings with detailed requirements and JD attachments, manage referrals, and monitor all employee activities. All these operations are accessible through clearly organized sections of the HR dashboard, making the day-to-day management of HR activities significantly more efficient than manual methods.

- **Employee Self-Service Capabilities:** The HRMS empowers employees with a comprehensive self-service portal. Employees can independently view their travel assignments, upload required travel documents, submit and track expense claims, participate in the social achievement feed, book game slots, refer candidates, and manage their profile — all without needing to contact HR directly for routine tasks.

This self-service model not only improves employee satisfaction but also significantly reduces the administrative burden on the HR team.

- **Manager Visibility and Team Monitoring:** Managers have a dedicated view that gives them read-only visibility into their team's activities. They can see which team members are assigned to travel, review the travel documents and expense submissions of their direct reports, view the organizational hierarchy, and monitor the social feed. This operational transparency enables managers to stay informed about their team's movements and activities without requiring separate communication or manual reporting.

- **Automated Notifications Reducing Communication Overhead:** One of the most operationally significant features of the HRMS is its automated notification system. Every key event in the system from travel assignment to expense approval, from referral submission to post moderation triggers both an in-app notification and an email to the relevant stakeholders. This eliminates the need for manual follow-up communication, ensures that no update is missed, and creates a reliable communication channel between employees and HR.

- **Automated Validation Reducing Human Error:** The system enforces strict validation at both the frontend and backend levels. Form fields are validated for correct formats, required fields are enforced, business rules such as expense date windows are automatically checked, and file uploads are validated for type and size. This automated validation significantly reduces the number of errors that would otherwise require manual correction, improving the overall quality of data in the system and reducing rework for both employees and HR.

6.1.3 Economical Feasibility :

The economical feasibility study evaluated the cost-effectiveness of building and deploying the HRMS. The analysis confirms that the system is highly economically feasible.

- **Open-Source and Freely Available Technologies:** The entire technology stack used in the HRMS ASP.NET Core, React.js, TypeScript, and Entity Framework Core is open-source and freely available. SQL Server is available in a free Developer or Express edition suitable for development and small to medium production environments. This means the organization incurs zero licensing costs for the development tools and frameworks, significantly reducing the overall cost of building the system compared to proprietary alternatives.

- **Significant Reduction in Operational Costs:** By automating travel management, expense tracking, approval workflows, job referrals, and game slot scheduling, the HRMS eliminates a large volume of manual administrative work. HR staff who previously spent hours every week processing paper forms, chasing approvals via email, and manually updating spreadsheets can now focus on more strategic HR activities. This reallocation of human effort represents a substantial reduction in operational costs over time.

- **Centralized Data Management Reducing Redundancy: The** HRMS consolidates all HR data into a single, well-structured SQL Server database. This eliminates the data redundancy, duplication, and inconsistency that arise when information is stored across multiple disconnected systems. Centralized data management also makes reporting and analytics significantly easier, further reducing the time and cost associated with generating management reports.

- **Low Maintenance Cost Due to Modular Architecture:** The HRMS is built with a clean, modular architecture where each functional area travel, expenses, posts, games, jobs is implemented as a separate module with its own controllers, services, and data access layers. This modularity means that future changes or bug fixes in one module do not affect others, reducing the time and cost of maintenance. New features can be added as independent modules without requiring changes to existing code.

- **Flexible and Low-Cost Deployment Options:** The HRMS can be deployed on a variety of environments depending on the organization's budget and preferences. For small organizations, it can run on a single low-cost cloud virtual machine or even on-premise hardware. For larger organizations requiring higher availability and scalability, it can be containerized using Docker and deployed on cloud platforms such as Azure or AWS with auto-scaling capabilities. This flexibility ensures that the deployment cost can be matched to the organization's current scale and budget.

- **Future Scalability Without Significant Additional Investment:** The modular architecture and cloud-ready design of the HRMS mean that the system can scale to accommodate more users, more modules, and higher data volumes without requiring a complete rebuild. New modules such as payroll management, leave management, or performance appraisal can be added incrementally. Additional users can be onboarded by simply provisioning more server resources, which in cloud environments is done on-demand and billed only for what is used.

6.2 System Requirement Specification :

6.2.1 Introduction :

The Human Resource Management System (HRMS) is an internal enterprise web-based application designed to automate and manage core HR operations such as travel management, expense reimbursement, employee engagement, games scheduling, organizational hierarchy visualization, and job referral workflows.

The system aims to reduce manual HR processes, improve transparency, and enhance employee engagement by providing a centralized digital platform accessible to Employees, Managers, and HR personnel.

I. User & Role Management Module
II. Travel & Expense Management Module
III. Social Achievements & Celebrations Module
IV. Games Scheduling Module
V. Organization Chart Module
VI. Job Listing, Sharing & Referral Module

6.2.2 Purpose :

The primary purpose of the HRMS is to provide a secure, centralized, and efficient platform for managing internal HR operations.

The system aims to:

- Automate travel assignments and expense reimbursements
- Reduce manual document handling
- Improve employee engagement through social features
- Ensure fair allocation of shared organizational resources (games)
- Provide transparency in organizational hierarchy
- Simplify job sharing and referral processes

The system helps HR teams:

- Manage employee travel plans and documents
- Track and verify expenses
- Configure job postings and referral workflows
- View organization structure across departments

The system enables employees to:

- View assigned travels and submit expenses
- Upload and manage travel-related documents
- Participate in internal social activities
- Book game slots fairly
- Refer friends for job openings

6.2.3 Problem Statement :

Organizations relying on manual or partially automated HR processes face challenges such as:

- Delays in travel communication and expense reimbursement
- Poor visibility of organizational hierarchy
- Unfair usage of shared recreational resources
- Inefficient job referral tracking
- Lack of employee engagement platforms

The HRMS solves these problems by digitizing and automating HR workflows with role-based access control and centralized data management.

6.2.4 Scope :

6.2.4.1 User Management Module

• **Registration/Login :** Employees, Managers, and HR users can securely log in to the system using authentication credentials.

• **Profile Management :** Users can view and manage their personal profile information within the system.

• **Role-Based Access :** The system supports different roles such as Employee, Manager, and HR, and each role can access features according to their permissions.

6.2.4.2 Travel Management Module

• **Travel Creation and Assignment :** HR can create travel plans and assign them to one or more employees.

• **View Travel Details :**
Employees can view their assigned travel plans and related information.

6.2.4.3 Expense Management Module

• **Expense Submission :** Employees can submit expenses related to their assigned travel with proof documents.

• **Expense Verification :** HR can verify, approve, or reject submitted expenses and update their status.

6.2.4.4 Achievements & Engagement Module

• **Social Achievements Feed :** Employees can create achievement posts and view posts shared by others.

• **Likes and Comments :** Users can like and comment on achievement posts and participate in engagement activities.

6.2.4.5 Game Slot Management Module

• **Game Slot Scheduling :** HR can configure game slots, including timing, duration, and capacity.

• **Fair Slot Allocation :** Employees can request game slots, and the system allocates them fairly according to defined rules.

6.2.4.6 Organization Chart Module

• **Hierarchy Visualization :** Employees can view the organizational hierarchy and managerial chain within the system.

• **Employee Relationship View :** The system displays reporting relationships between employees and managers.

6.2.4.7 Job Listing & Referral Module

• **Job Listing and Sharing :** HR can create and manage job openings available in the organization.

• **Employee Referrals :** Employees can refer candidates for job openings and track referral status.

6.2.5 Audience and Users :

I. Employee: Views travels, submits expenses, posts achievements, books game slots, refers jobs

II. Manager: View team member, travel documents, expense (read-only), organizational chart

III. Hr: Manager travels, expenses, documents, jobs, referrals, and organizational chart

6.2.6 Overview :

6.2.6.1 Hardware and software (for development)

Hardware Requirements	
Processor	Intel core i5 or higher
RAM	8 GB
Software Requirements	
Backend	ASP.NET Core
Database	SQL Server
Tools	Visual Studio, Visual Studio code, Postman
Browser	Latest versions of Chrome, Edge, Firefox

6.2.6.2 Hardware and software (for User)

Hardware Requirements	
Processor	Intel core i3 or higher
RAM	4 GB
Software Requirements	
Operating System	Windows, Ios, Android
Browser	All Latest Browser

6.2.7 Acronyms and Abbreviations :

- HRMS – Human Resource Management System
- API – Application Programming Interface
- DB – Database
- JWT – JSON Web Token
- RBAC – Role-Based Access Control

6.2.8 Functional Requirements :

6.2.8.1 User & Role Management module :

a. Common Requirements :

The authentication system is the gateway to the entire HRMS. Every user whether an Employee, Manager, or HR Administrator must authenticate using their registered email address and a secure password before gaining access to any part of the system. Passwords are stored in the database in a hashed format using a strong hashing algorithm, ensuring that even if the database is compromised, raw passwords cannot be retrieved. The system issues a JWT token upon successful login, which is included in the header of every subsequent API request. This token has a defined expiry duration, after which the user must log in again to obtain a new token, ensuring that sessions do not remain active indefinitely.

Role-based access control is enforced at both the API level and the UI level. At the API level, each endpoint is decorated with authorization attributes that specify which roles are permitted to call it. At the UI level, navigation menus and page components are conditionally rendered based on the logged-in user's role. This dual enforcement ensures that even if a user manually tries to navigate to a restricted URL, the backend API will reject the request.

- The system shall authenticate users using secure login credentials (email and password).
- The system shall enforce role-based access control at both API and UI levels.
- The system shall store passwords in hashed format using a secure hashing algorithm.
- The system shall issue a JWT token with a defined expiry upon successful login.
- The system shall reject API requests that do not include a valid, non-expired JWT token.

b. Employee :

Employees are the primary end-users of the HRMS. Each employee has a profile that includes their full name, email address, profile picture, designation, department, date of birth, date of joining, and the manager they report to. Employees can view and update certain personal profile information through the self-service portal. They are linked to a specific department and report to a designated manager, which determines how they appear in the organizational chart and whose approval chain they fall under for certain workflows.

- The employee shall be able to view and update their personal profile information including name, profile picture, and designation.
- The employee shall be assigned to a department and shall report to a designated manager.
- The employee's reporting structure shall be reflected accurately in the organizational chart.
- The employee shall receive in-app and email notifications for all key events related to their activities.

c. Manager :

Managers occupy a middle tier in the organizational hierarchy. They have all the capabilities of a regular employee but additionally have read-only visibility into the activities of their direct reports. A manager can view which employees report to them, monitor their travel assignments and expense submissions, and see them represented in the organizational chart. Managers do not have the ability to approve or reject expenses that authority belongs to HR but their visibility into team activities makes them effective coordinators.

- The manager shall be able to view a list of all their direct reports.
- The manager shall have read-only access to the travel plans and expense submissions of their direct reports.
- The manager shall appear correctly in the organizational chart as a reporting node above their team members.
- The manager shall be able to view team members' travel documents without the ability to modify them.

d. HR :

HR Administrators have the highest level of access in the system. They are responsible for managing the complete lifecycle of all HR operations. HR can create and manage employee records, assign travel plans, verify and update expense statuses, configure game slot rules, publish job openings, review referrals, and moderate social posts. Because HR has access to sensitive employee data such as salaries, documents, and performance records, their actions are subject to the strictest audit logging in the system.

- HR shall be able to view, create, update, and soft-delete employee records.
- HR shall manage the complete lifecycle of travel plans from creation to closure.
- HR shall verify and update the status of submitted expense claims with mandatory remarks on rejection.
- HR shall configure game settings including operating hours, slot durations, and player limits.
- HR shall create and manage job openings including uploading job description documents.
- HR shall review and update the status of job referrals submitted by employees.
- HR shall have the ability to mark social posts as inappropriate and notify the post author via email.

6.2.8.2 Travel & Expense Management Module :

a. Travel Assignment:

The travel assignment process begins when an HR Administrator identifies the need to send one or more employees on an official trip. HR logs into the system and creates a travel plan by filling in details such as the travel title, a description of the purpose of the trip, the destination location, the start date, the end date, and the maximum expense amount that each traveller is permitted to claim. Once created, HR assigns one or more employees to the travel plan. Each assigned employee immediately receives an in-app notification and an email alert containing the travel details, so they can plan accordingly.

The travel plan serves as the parent record for all subsequent activities related to that trip document uploads, expense submissions, and expense approvals are all linked back to the specific travel plan. This creates a clean, auditable record of everything related to each official trip.

- HR shall be able to create travel plans with title, description, location, start date, end date, and maximum expense limit.
- HR shall be able to assign one or more employees to a single travel plan.
- The system shall send in-app and email notifications to all assigned employees immediately upon assignment.
- Each travel plan shall serve as the parent record for all related documents and expense submissions.
- Employees shall be able to view the complete details of their assigned travel plans at any time

b. Travel Document:

Travel documents are a critical part of the travel management workflow. Both HR and employees can upload documents related to a specific travel plan. These documents may include flight tickets, hotel booking confirmations, visa copies, meeting agendas, and any other relevant records. Each uploaded document is stored with metadata including the travel it belongs to, the employee it relates to, the user who uploaded it, the document type, the file URL, and the upload timestamp.

Access to travel documents is role-controlled. HR can view and upload documents for all employees across all travel plans. Employees can only view and upload documents related to their own assigned travels. Managers have read-only access to the travel documents of their direct reports, giving them visibility without the ability to modify records.

- The system shall allow both HR and employees to upload documents for specific travel plans.
- Each document record shall store Travel ID, Employee ID, Uploaded By, Document Type, File URL, and Upload Timestamp.

- HR shall have view and upload access to all travel documents across all employees.
- Employees shall have view and upload access only to documents related to their own assigned travels.
- Managers shall have read-only access to travel documents of their direct reports.
- Uploaded files shall be validated for acceptable file types and sizes before being saved.

c. Expense Submission:

The expense submission module allows employees to claim reimbursement for costs incurred during official travel. Every expense submission must be linked to a specific travel plan standalone expense submissions not associated with a travel are not permitted. This linkage ensures that all claimed expenses are traceable to a legitimate organizational trip.
To prevent fraudulent or late submissions, the system enforces a strict submission window. Employees may only submit expenses after the travel start date has passed, and all submissions must be made within 10 days of the travel end date. After this window closes, the system automatically blocks any further submissions for that travel. Multiple expense submissions per travel are supported, as employees may incur several different types of expenses during a single trip such as transportation, accommodation, meals, and miscellaneous costs.

- All expense submissions shall be linked to a specific travel plan.
- Employees shall only be able to submit expenses after the travel start date.
- The system shall automatically block expense submissions made more than 10 days after the travel end date.
- Multiple expense submissions per travel plan shall be supported.
- Each expense record shall capture the amount, category, date, details, and current approval status.

d. Proof Upload and Verification:

Every expense submission must be accompanied by at least one proof document such as a scanned receipt, invoice, or bill. The system enforces this requirement at the point of submission an expense cannot be submitted without at least one attached proof document. This mandatory proof requirement significantly reduces the risk of fraudulent or inflated expense claims.

Once an employee submits their expense, the status changes from Draft to Submitted and the HR Administrator is notified. HR can then open the expense record, review the submitted proof documents, and either approve or reject the claim. If HR rejects the expense, they must provide a mandatory rejection remark explaining the reason. The employee is notified of the decision via in-app

notification and email, and they can view the remark to understand why the claim was rejected.

- Each expense submission shall require at least one proof document to be attached before submission.
- The system shall prevent expense submission if no proof document is attached.
- HR shall be able to review all submitted proof documents before making an approval decision.
- HR shall be able to approve or reject expense claims with a mandatory remark on rejection.
- The system shall notify employees via in-app and email notifications upon approval or rejection.
- Expense status shall progress through the following states: Draft → Submitted → Approved / Rejected.

6.2.8.3 Social Achievement & Celebrations Module

a. Achievement Posts:

The social achievement module serves as an internal social feed for the organization, designed to recognize and celebrate employee accomplishments, project milestones, team successes, and personal achievements. Employees can create posts by providing a title, a detailed description, relevant tags from a predefined list, an image attachment, and a visibility setting (public or private). Public posts are visible to all employees in the organization, while private posts are only visible to the employee themselves.
All posts are displayed in the feed in reverse chronological order, so the most recent content always appears at the top. Any logged-in user can interact with posts by liking them or adding comments, fostering a sense of community and mutual recognition within the organization. This social engagement layer is a key part of the employee engagement strategy embedded in the HRMS.

- Employees shall be able to create achievement posts with a title, description, image, tags, and visibility setting.
- All users shall be able to view public posts in reverse chronological order on the social feed.
- Users shall be able to like posts, with the like count displayed on each post.
- Users shall be able to comment on posts, with all comments displayed under the post in chronological order.
- Employees shall be able to edit or delete their own posts at any time.

b. Moderation:

To maintain a professional and respectful environment in the social feed, HR Administrators have the ability to moderate posts. If HR identifies a post that contains inappropriate, offensive, or unprofessional content, they can mark the post as inappropriate, which hides it from the general feed. When a post is marked as inappropriate, the system automatically sends a warning email to the author explaining that their post has been removed due to a policy violation.

This moderation capability ensures that the social feed remains a positive space for employee recognition and engagement, while giving HR the tools to enforce the organization's communication policies.

- HR shall be able to mark any post as inappropriate, removing it from the public feed.
- The system shall automatically send a warning email to the post author when their post is marked as inappropriate.
- Employees shall only be able to edit or delete their own posts; they cannot moderate others' posts.
- HR shall have a dedicated post management page listing all posts with the ability to filter by status.

c. Birthday & Anniversary:

The system includes an automated celebration feature that generates system posts for two types of events employee birthdays and work anniversaries. The system checks the date of birth and date of joining fields for all active employees daily. When a birthday or work anniversary is detected, the system automatically creates a celebration post visible to all users in the organization.

This automated recognition feature ensures that no employee's special day goes unnoticed, fostering a culture of appreciation and belonging within the organization. The system posts are generated without any manual intervention from HR, reducing their administrative workload while still delivering a meaningful employee experience.

- The system shall automatically check employee dates of birth and dates of joining on a daily basis.
- The system shall auto-generate a birthday post for each employee on their date of birth.
- The system shall auto-generate a work anniversary post for each employee on their joining anniversary.
- All auto-generated celebration posts shall be visible to all users in the organization.
- Auto-generated posts shall be clearly marked as system-generated to distinguish them from employee-created content.

6.2.8.4 Game Scheduling Module :

a. Configuration:

The game scheduling module is designed to manage the fair usage of shared recreational facilities within the organization such as table tennis, foosball, or carrom. HR Administrators are responsible for configuring each game in the system. The configuration includes specifying the game name, the minimum and maximum number of players per slot, the duration of each slot in minutes, the buffer time before a slot is finalized (slot assigned before X minutes), and how many days in advance slots should be auto-generated.

Once a game is configured, the system automatically generates daily slots based on the operating hours and slot duration. For example, if table tennis operates from 12:00 PM to 2:00 PM and each slot is 30 minutes long, the system will automatically create four slots per day. This eliminates the need for HR to manually create slots every day.

- HR shall be able to create and configure games with name, min/max players, slot duration, buffer time, and advance slot generation days.
- HR shall be able to configure operating hours for each game through the game operation window settings.
- The system shall automatically generate daily game slots based on the configured operating hours and slot duration.
- HR shall be able to update game configurations, with changes reflected in future auto-generated slots.

b. Booking Rules:

Employees can request game slots by browsing the available slots for each game and submitting a booking request. The system enforces a rule of one active booking per employee per day across all games, ensuring that no single employee can monopolize the available slots. Fair scheduling is the central principle of this module employees who have played less recently are given higher priority in the queue.

If a slot is fully booked, employees can join a waiting list. When a booked employee cancels their slot, the system automatically assigns it to the next eligible player on the waiting list based on the fair scheduling algorithm, and notifies them immediately. This ensures that cancellations do not result in wasted slot capacity.

- Employees shall be able to browse available game slots and submit booking requests.
- The system shall enforce a maximum of one active booking per employee per day.
- Fair scheduling shall be enforced, prioritizing employees who have played less recently.
- Employees shall be able to join a waiting list if all slots for a game are fully booked.
- Upon cancellation by a booked player, the system shall automatically assign the slot to the next eligible player on the waiting list.

6.2.8.5 Organization Chart Module :

The organization chart module provides a visual representation of the company's reporting hierarchy. It is accessible to all logged-in users and serves as a reference for understanding the organizational structure. Each employee is represented as a node in the chart, displaying their name, designation, and profile picture.
The chart is interactive clicking on any node reloads the chart centered on that employee, showing their manager above and their direct reports below. This makes it easy to navigate through the entire organizational hierarchy from any starting point. The chart is dynamically generated from the reporting relationships stored in the users table, so it always reflects the current structure without requiring any manual updates.

- The system shall display a dynamic organizational chart based on current reporting relationships.
- Each node in the chart shall display the employee's name, designation, and profile picture.
- The chart shall show the selected employee's manager above and direct reports below.
- Clicking on any node shall reload the chart centered on the selected employee.
- The chart shall be accessible to all logged-in users regardless of role.
- The chart shall automatically reflect any changes to reporting relationships without manual intervention.

6.2.8.6 Job Listing, Sharing & Referral Module :

a. Job Listing:

HR Administrators can create and publish job openings within the organization. Each job listing includes the job title, role, location, requirements, and an attached job description PDF document. Job listings are visible to all employees in the system, allowing them to stay informed about current openings. Employees can browse job details to understand the requirements and decide whether they know suitable candidates to refer.

- HR shall be able to create job listings with title, role, location, requirements, and a job description PDF.
- HR shall be able to assign one or more HR contacts and reviewers to each job listing.
- All active job listings shall be visible to all logged-in employees.
- Employees shall be able to view the complete details of each job listing including the attached job description document.
- HR shall be able to deactivate job listings that are no longer open, hiding them from the employee view.

b. Job Sharing:

Employees who find a job listing relevant to someone in their network can share the job directly via email through the system. The system automatically attaches the job description PDF to the email and sends it to the specified recipient's email address. All sharing activities are logged in the database with the employee who shared, the recipient email, the job shared, and the timestamp. This logging creates an audit trail of outreach activities for each job opening.

- Employees shall be able to share any active job listing via email directly through the system.
- The system shall automatically attach the job description PDF to the sharing email.
- All job sharing activities shall be logged with the sharing employee, recipient email, job, and timestamp.
- Employees shall be able to share the same job to multiple recipients in separate sharing actions.

c. Job Referrals:

The job referral module allows employees to formally recommend candidates for open positions. When submitting a referral, the employee must provide the candidate's full name, email address, an uploaded CV, and an optional personal note. Once submitted, the system automatically sends an email to the HR contact assigned to the job and all configured reviewers, notifying them of the new referral and attaching the candidate's CV.
The referral status is tracked throughout the hiring process, allowing the referring employee to monitor what stage their referral is at. This transparency motivates employees to make quality referrals and stay engaged in the recruitment process.

- Employees shall be able to submit referrals for any active job listing with the candidate's name, email, CV, and note.
- The system shall automatically notify the HR contact and all configured reviewers via email upon referral submission.
- The referral status shall be tracked and updated by HR reviewers throughout the recruitment process.
- Employees shall be able to view the current status of all their submitted referrals.
- The system shall validate that the candidate's CV is uploaded in an acceptable file format before submission.

6.2.9 Non-Functional Requirements :

Non-functional requirements define the quality attributes and operational standards that the HR Management System must meet. While functional requirements describe **what** the system does, non-functional requirements describe **how well** the system performs its functions. These requirements are equally critical to the success of the system because even a functionally complete system will fail in a real-world enterprise environment if it is slow, insecure, unavailable, or difficult to maintain. The following non-functional requirements have been identified and defined for the HRMS.

1. Accessibility

Accessibility refers to the ability of users to access and use the system regardless of the device, browser, or screen size they are using. In a modern workplace, employees may need to access HR features from a desktop computer in the office, a laptop at home, or a mobile phone while travelling. The HRMS must accommodate all of these scenarios without requiring users to install any dedicated software.

The system is built as a web-based application using React.js with TypeScript, which enables it to run in any modern browser without plugins or installations. The UI is developed using responsive design principles, meaning the layout automatically adapts to different screen sizes and resolutions. Whether an employee is viewing the system on a 27-inch desktop monitor or a 5-inch mobile screen, the interface will reorganize itself to remain usable and readable.

All interactive elements such as buttons, forms, dropdowns, and navigation menus are designed to be clearly visible, properly labeled, and easy to interact with on both mouse-driven and touch-driven devices. Form validation messages are displayed clearly next to the relevant fields so users immediately understand what correction is needed. Error states, loading indicators, and success confirmations are all communicated through clear visual feedback, ensuring that users always know the current state of their actions.

- The system shall be accessible via all modern web browsers including Chrome, Edge, Firefox, and Safari without requiring any plugins or additional software installation.
- The UI shall be fully responsive, automatically adjusting its layout to provide an optimal experience across desktop, tablet, and mobile screen sizes.
- All interactive elements shall be clearly labeled and easy to interact with on both mouse-driven and touch-driven devices.
- Form validation messages and system feedback shall be clearly displayed to guide users through correct usage of the system.
- The system shall not require any browser-specific features that limit compatibility to a single browser.

2. Availability

Availability refers to the proportion of time that the system is operational and accessible to its users. In an enterprise environment, HR operations do not always happen within standard office hours. Employees may submit expense claims late at night, managers may review team activities on weekends, and HR may need to create travel plans outside of business hours. The HRMS must therefore be available around the clock to support these use cases.

The system is designed to achieve 24×7 availability except during planned maintenance windows, which should be scheduled during off-peak hours and communicated to users in advance. High availability is achieved through a combination of reliable infrastructure, proper error handling, and graceful degradation meaning that even if a non-critical component fails, the rest of the system continues to function normally.

The system is architected to support multiple concurrent users without degradation in performance or reliability. Connection pooling is used at the database level to efficiently manage simultaneous database connections. The backend API is stateless, meaning each request is independent and the server does not need to maintain session state between requests, which makes horizontal scaling straightforward when additional capacity is needed.

Planned maintenance activities such as database backups, software updates, and infrastructure upgrades are designed to be performed with minimal disruption. Where possible, rolling updates are used so that the system remains partially or fully available even during deployment of new versions.

- The system shall be available 24 hours a day, 7 days a week, except during planned maintenance windows.
- Planned maintenance windows shall be scheduled during off-peak hours and communicated to users in advance.
- The system shall support multiple concurrent users without performance degradation or data integrity issues.
- Connection pooling shall be implemented at the database level to efficiently manage simultaneous database connections.
- The backend API shall be stateless to support horizontal scaling when additional capacity is required.
- The system shall implement graceful error handling so that failures in non-critical components do not bring down the entire application.
- Database backups shall be performed regularly to ensure data recovery in the event of a failure.

3. Performance

Performance refers to how quickly and efficiently the system responds to user actions. In an enterprise HR system used by potentially hundreds of employees simultaneously, slow response times lead to frustration, reduced productivity, and ultimately abandonment of the system in favor of manual processes. The HRMS is therefore designed with performance as a first-class concern at every layer of the architecture.

At the database level, all frequently queried tables are properly indexed to ensure fast data retrieval. For example, the users table is indexed on email for fast login lookups, and the expenses table is indexed on TravelId and TravelerId for fast expense retrieval per travel plan. Query optimization is applied across all modules to ensure that complex joins and aggregations are performed efficiently. Entity Framework Core queries are reviewed to avoid the N+1 query problem, where a separate database query is executed for each item in a list.

At the API level, response payloads are kept lean by returning only the data fields required by the frontend for each specific operation. This is achieved through the use of Data Transfer Objects (DTOs) that map database entities to lightweight response models. Pagination is implemented for all list-based endpoints such as the expense list, post feed, and job listings to ensure that the system never loads thousands of records at once, which would degrade both server and client performance.
At the frontend level, React's component-based architecture and virtual DOM ensure that UI updates are efficient and targeted, re-rendering only the components that have actually changed. Lazy loading is used for routes and large components so that the initial page load is fast and additional code is only downloaded when needed.

- All frequently queried database tables shall be properly indexed to ensure fast data retrieval.
- API responses shall return only the data fields required by the frontend, using DTOs to minimize payload size.
- Pagination shall be implemented on all list-based endpoints to prevent performance degradation when large amounts of data are present.
- The frontend shall use React's virtual DOM and lazy loading to ensure fast initial load times and efficient UI updates.
- Complex database queries involving joins and aggregations shall be optimized to execute within acceptable response time limits.
- The system shall avoid the N+1 query problem by using proper query strategies in Entity Framework Core.
- File uploads shall be processed asynchronously where possible to prevent blocking the main request thread and degrading API response times.

4. Security

Security is one of the most critical non-functional requirements for an enterprise HR system, as it handles highly sensitive data including employee personal information, financial expense data, confidential documents, CVs, and organizational hierarchy details. A security breach in such a system could have serious legal, financial, and reputational consequences for the organization. The HRMS implements a multi-layered security model to protect against a wide range of threats.
The first layer of security is authentication. Every user must log in with their registered email and password before accessing any part of the system. Passwords are never stored in plain text they are hashed using a strong, industry-standard hashing algorithm before being saved to the database. Even if the database were compromised, raw passwords would not be recoverable. Upon successful login, the system issues a JWT token that is signed with a secret key. This token must be included in the Authorization header of every subsequent API request.

The second layer is authorization. The JWT token contains the user's role, which is validated on every API request by the backend middleware. Each API endpoint specifies which roles are permitted to access it, and requests from unauthorized roles are rejected with an appropriate error response. This role-based authorization ensures that even authenticated users cannot access data or perform actions outside their permitted scope.
The third layer is input validation and sanitization. All data submitted by users form fields, file uploads, query parameters is validated on both the frontend and the backend. The backend validation is particularly important because it cannot be bypassed by a malicious user who submits requests directly to the API. File uploads are validated for acceptable MIME types and file sizes to prevent malicious file uploads. SQL injection is prevented by using Entity Framework Core's parameterized queries, which ensure that user input is never directly interpolated into SQL statements.

The fourth layer is transport security. All communication between the frontend and the backend API occurs over HTTPS, ensuring that data in transit is encrypted and cannot be intercepted or tampered with by a man-in-the-middle attacker.

- JWT-based authentication shall be enforced for all API endpoints, with tokens signed using a secure secret key.
- Passwords shall be stored in hashed format using a strong hashing algorithm; plain text passwords shall never be stored.
- Role-based authorization shall be applied at the API level, rejecting requests from users who do not have the required role.
- All user input shall be validated and sanitized on both the frontend and the backend to prevent injection attacks.
- File uploads shall be validated for acceptable MIME types and file sizes before being processed or stored.
- Entity Framework Core parameterized queries shall be used throughout to prevent SQL injection attacks.
- All client-server communication shall occur over HTTPS to ensure encryption of data in transit.
- JWT tokens shall have a defined expiry duration to prevent indefinitely active sessions.
- Sensitive data fields such as passwords shall never be included in API response payloads.

5. Maintainability

Maintainability refers to how easily the system can be understood, modified, extended, and debugged over its operational lifetime. Enterprise software systems are rarely static requirements change, bugs are discovered, new features are requested, and the technology landscape evolves. A system that is difficult to maintain becomes a liability over time, requiring disproportionate effort for even small changes. The HRMS is designed from the ground up with maintainability as a core architectural principle.

The backend follows a clean layered architecture with three distinct layers the Controller layer, the Service layer, and the Data Access layer. Controllers are responsible only for receiving HTTP requests and returning responses. All business logic is encapsulated in the Service layer, making it independently testable and reusable. The Data Access layer handles all interactions with the database through Entity Framework Core repositories. This separation of concerns ensures that changes to one layer do not cascade unnecessarily into others.

The frontend is built using React's component-based architecture, where the UI is broken down into small, reusable components. Each component is responsible for a single piece of the UI and its associated behavior. This modularity makes it easy to locate, understand, and modify specific parts of the UI without affecting unrelated components. TypeScript's static type checking catches type errors at compile time, reducing the number of runtime bugs and making the codebase easier to refactor confidently.

The database schema is managed through Entity Framework Core migrations, which provide a versioned history of all schema changes. Adding a new table or modifying an existing one is done through a migration, which can be applied to any environment consistently. This approach eliminates the risk of schema drift between development, testing, and production environments.

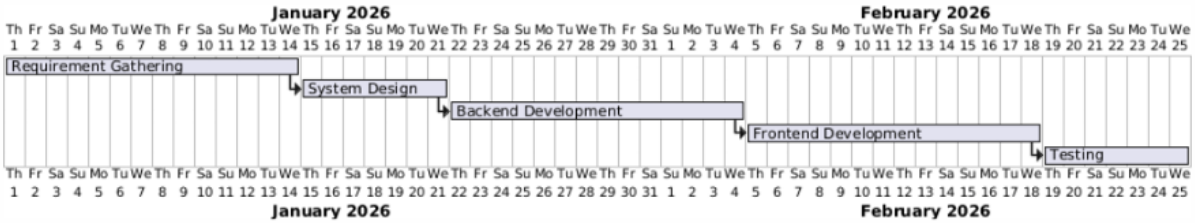
Code is written following established coding standards and conventions, with meaningful variable and function names, appropriate comments for complex logic, and consistent formatting. This makes the codebase readable and understandable for any developer who joins the team in the future.

- The backend shall follow a clean three-layer architecture Controller, Service, and Data Access with clear separation of concerns.
- All business logic shall be encapsulated in the Service layer to ensure it is independently testable and reusable.
- The frontend shall be built using React's component-based architecture, with each component responsible for a single, well-defined piece of the UI.
- TypeScript shall be used throughout the frontend to provide static type checking and reduce runtime errors.
- Database schema changes shall be managed through Entity Framework Core migrations to ensure consistent and versioned schema management across all environments.

- Code shall follow established naming conventions and coding standards to ensure readability and consistency.
- Complex business logic shall be documented with appropriate comments to aid future developers in understanding the implementation.
- The modular architecture shall ensure that new features can be added as independent modules without requiring modifications to existing code.
- Unit tests shall be written for critical service layer functions to ensure that future changes do not introduce regressions.

6.2.10 Other Information :

a. Gantt Chart :

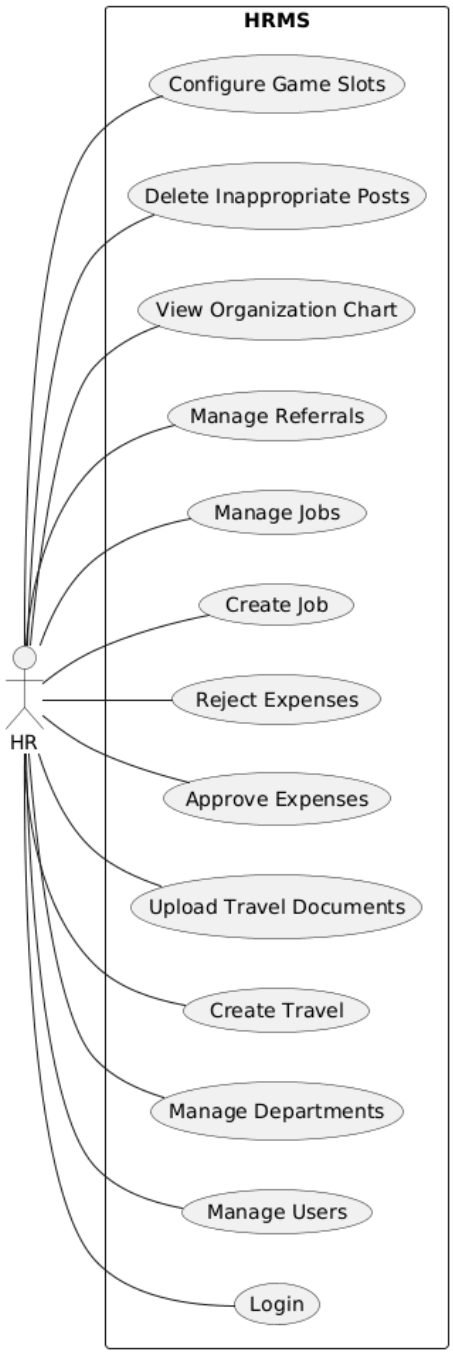


7. SYSTEM DESIGN

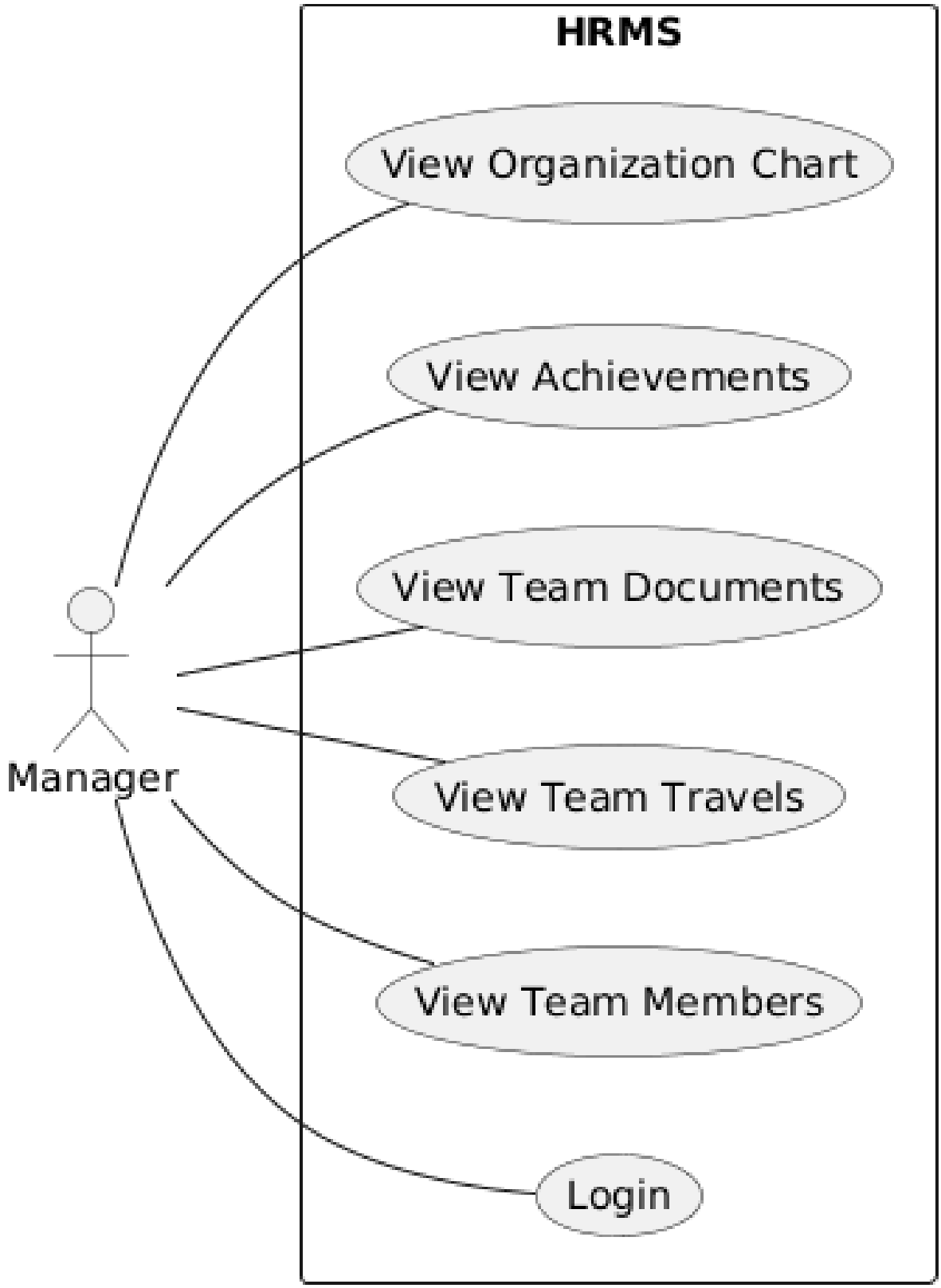
7.1 Architectural Design :

7.1.1 Usecase Diagrams :

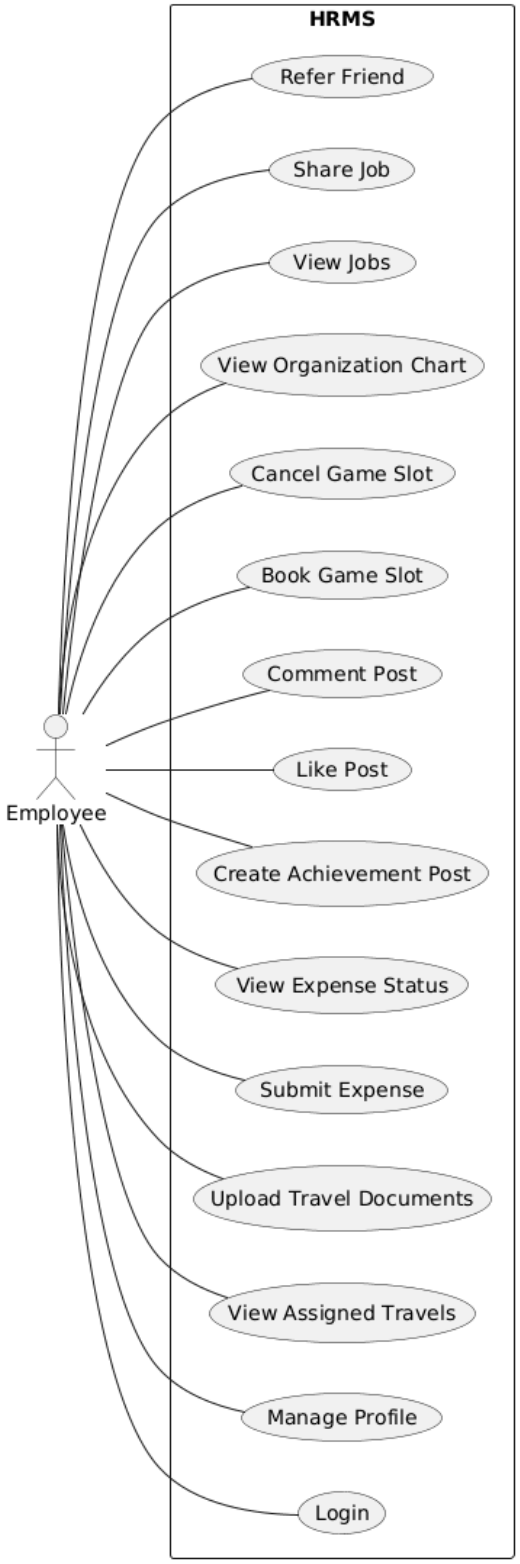
7.1.1.1 HR usecase Diagram:



7.1.1.2 Manager usecase Diagram:

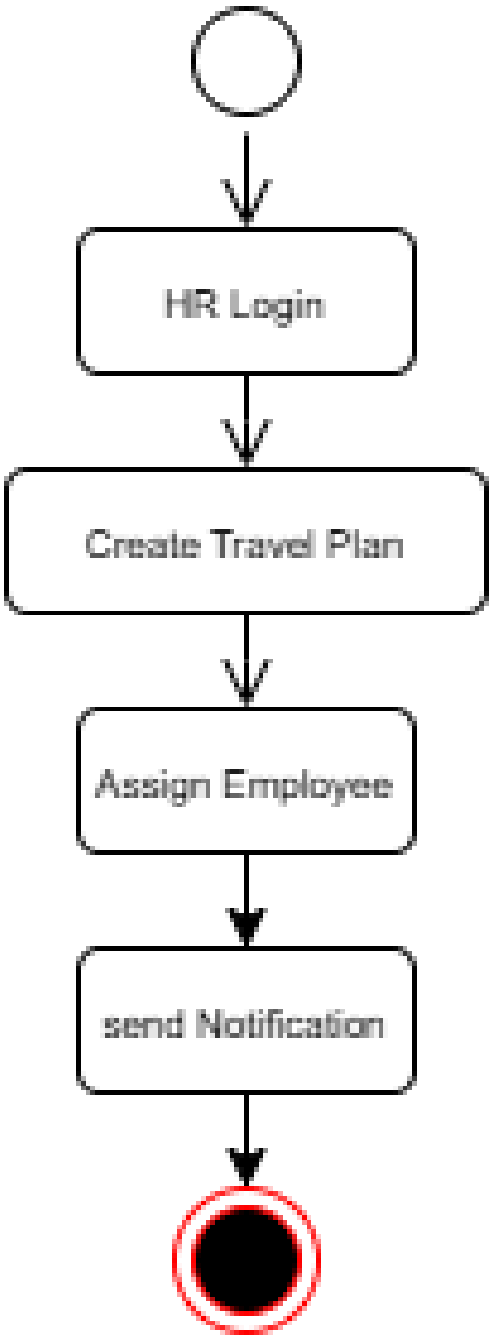


7.1.1.3 Employee usecase Diagram:

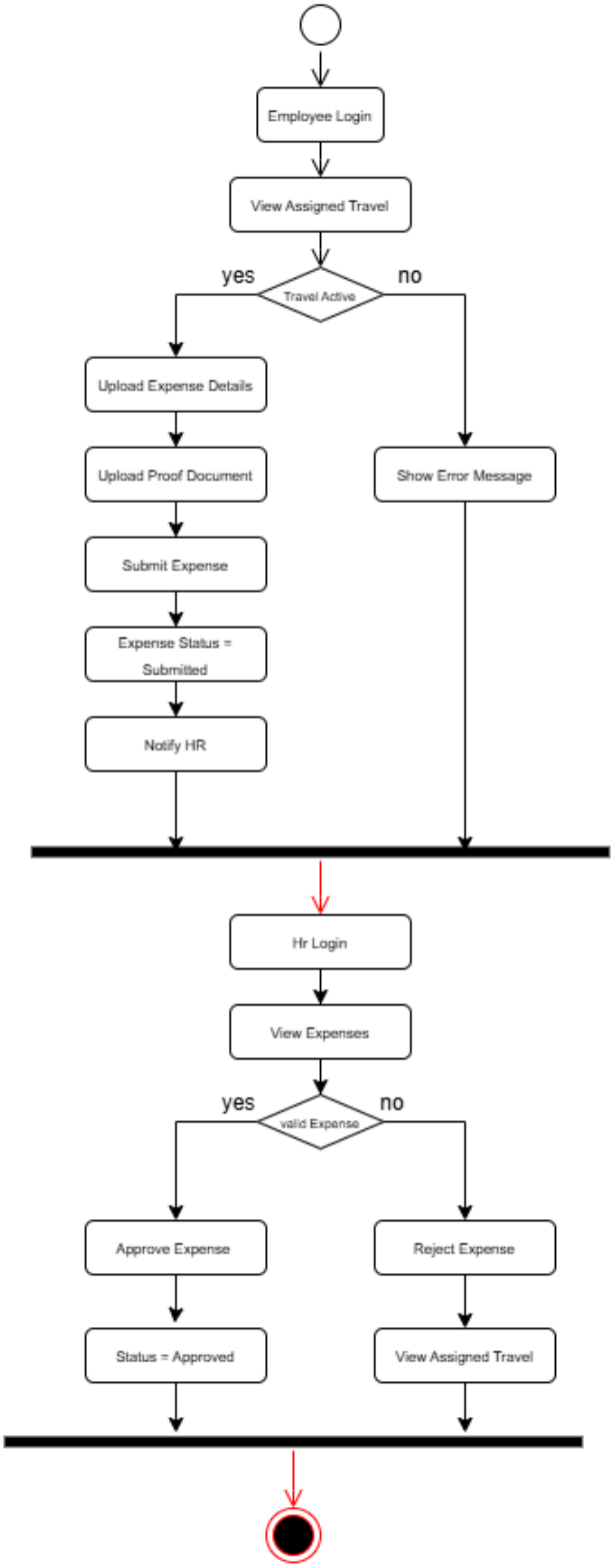


7.1.2 Activity Diagrams :

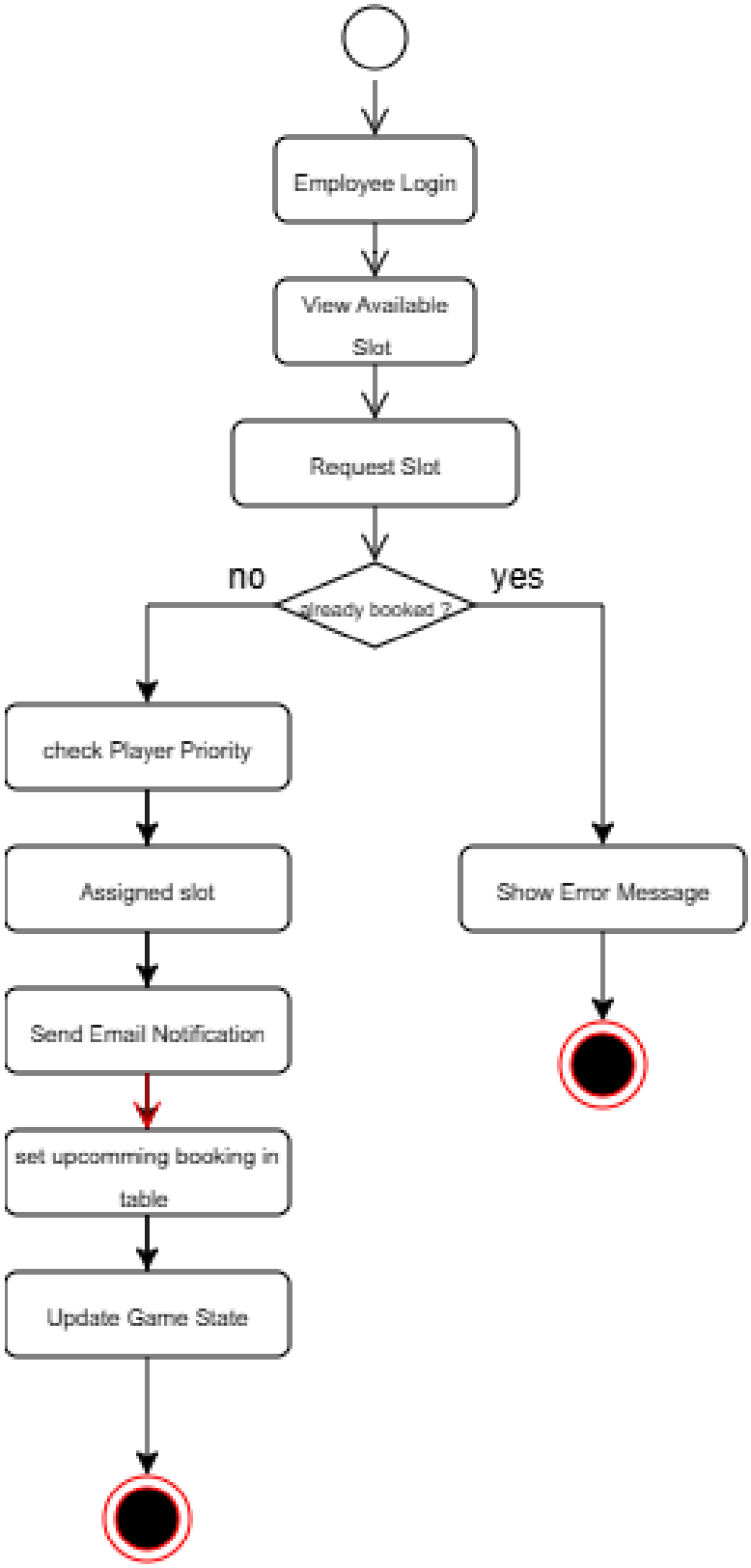
7.1.2.1 Create Travel :



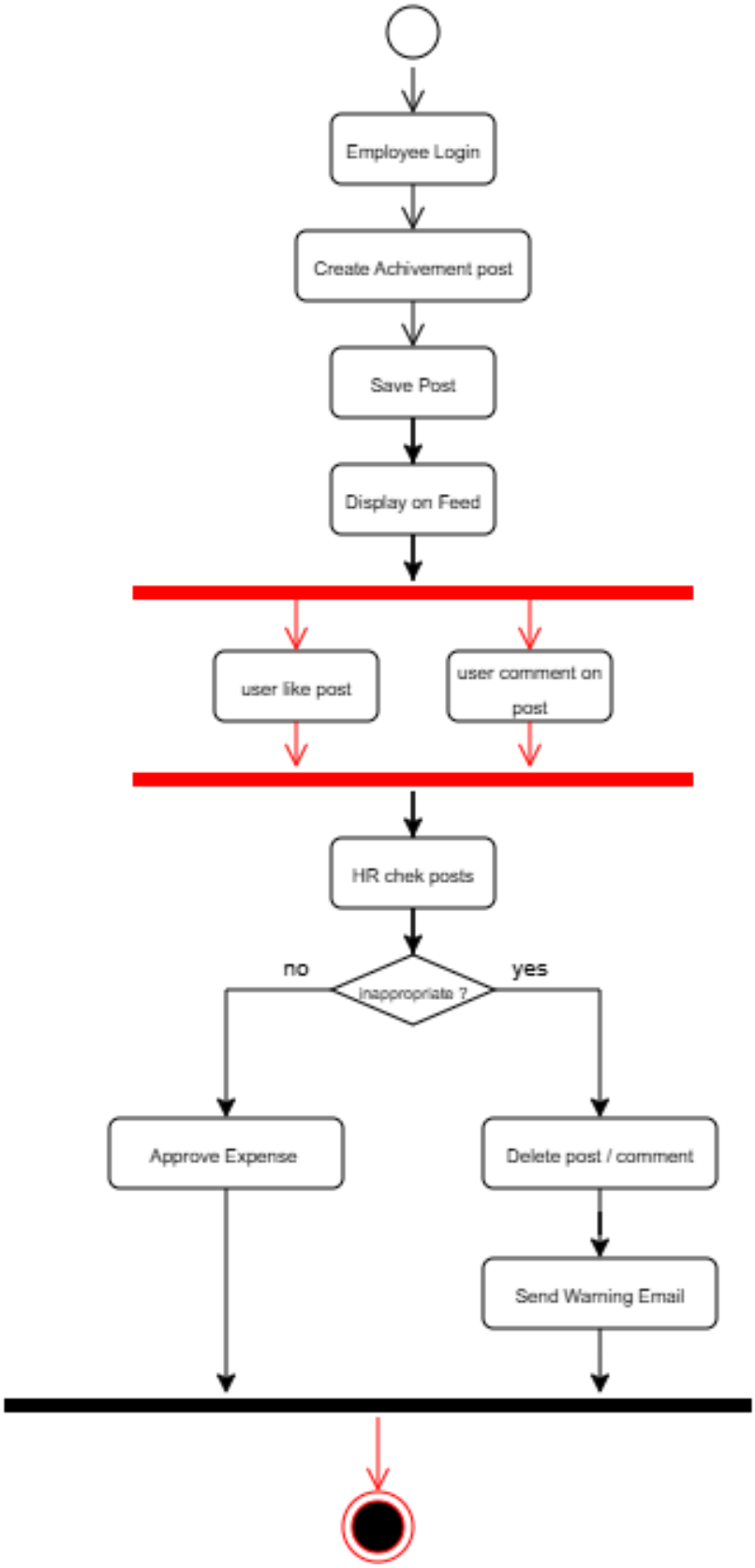
7.1.2.2 Travel Expense :



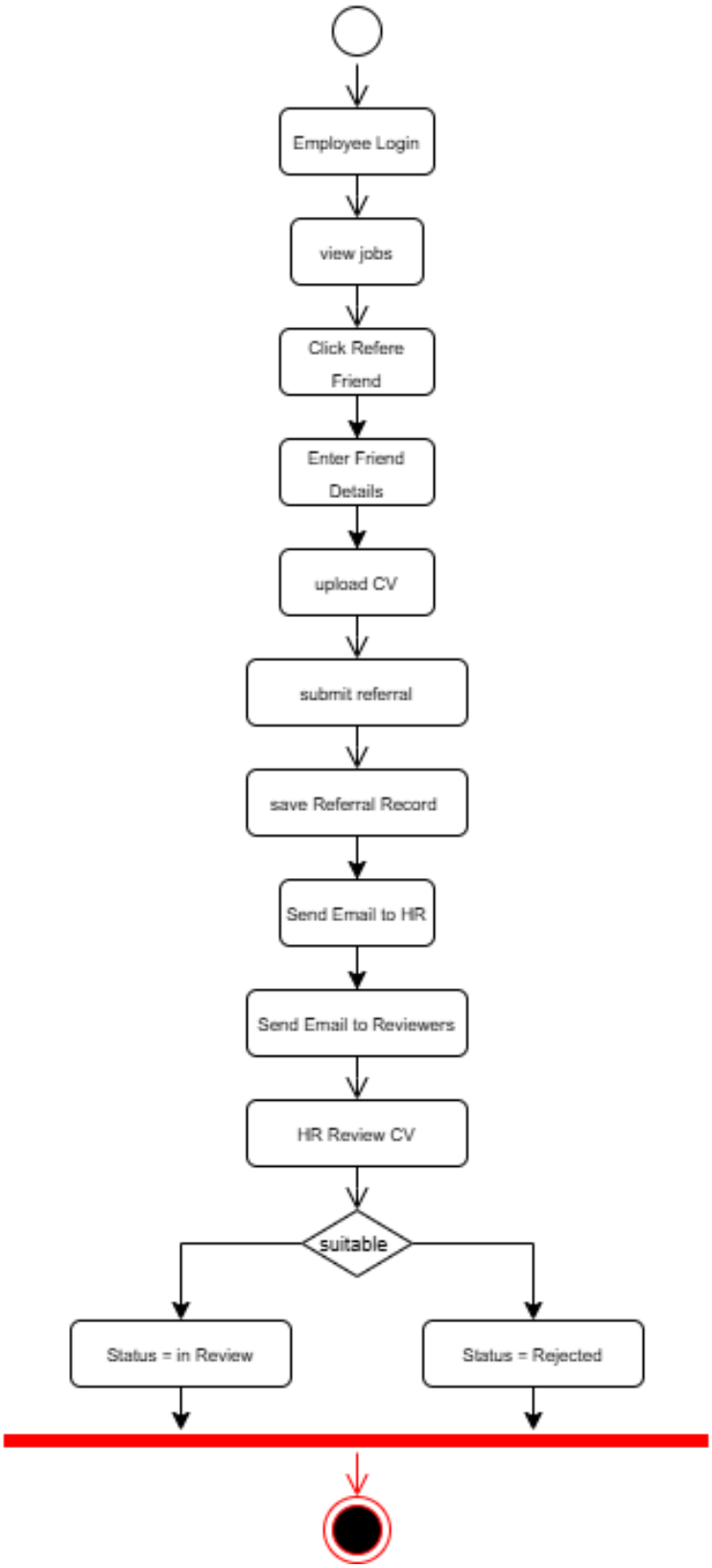
7.1.2.3 Book Game :



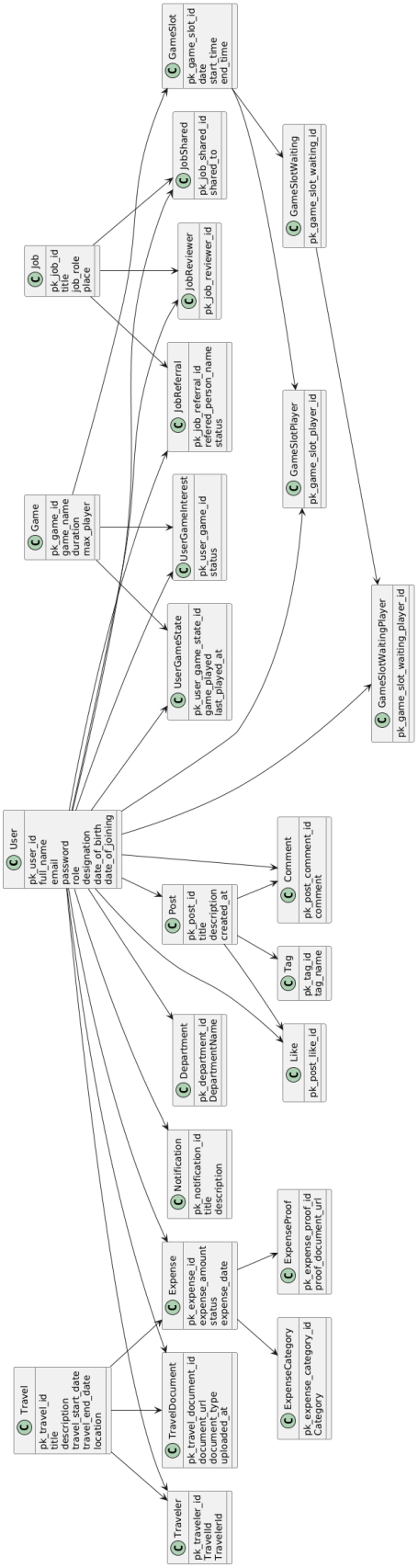
7.1.2.4 Post Create :



7.1.2.5 Job Referral :

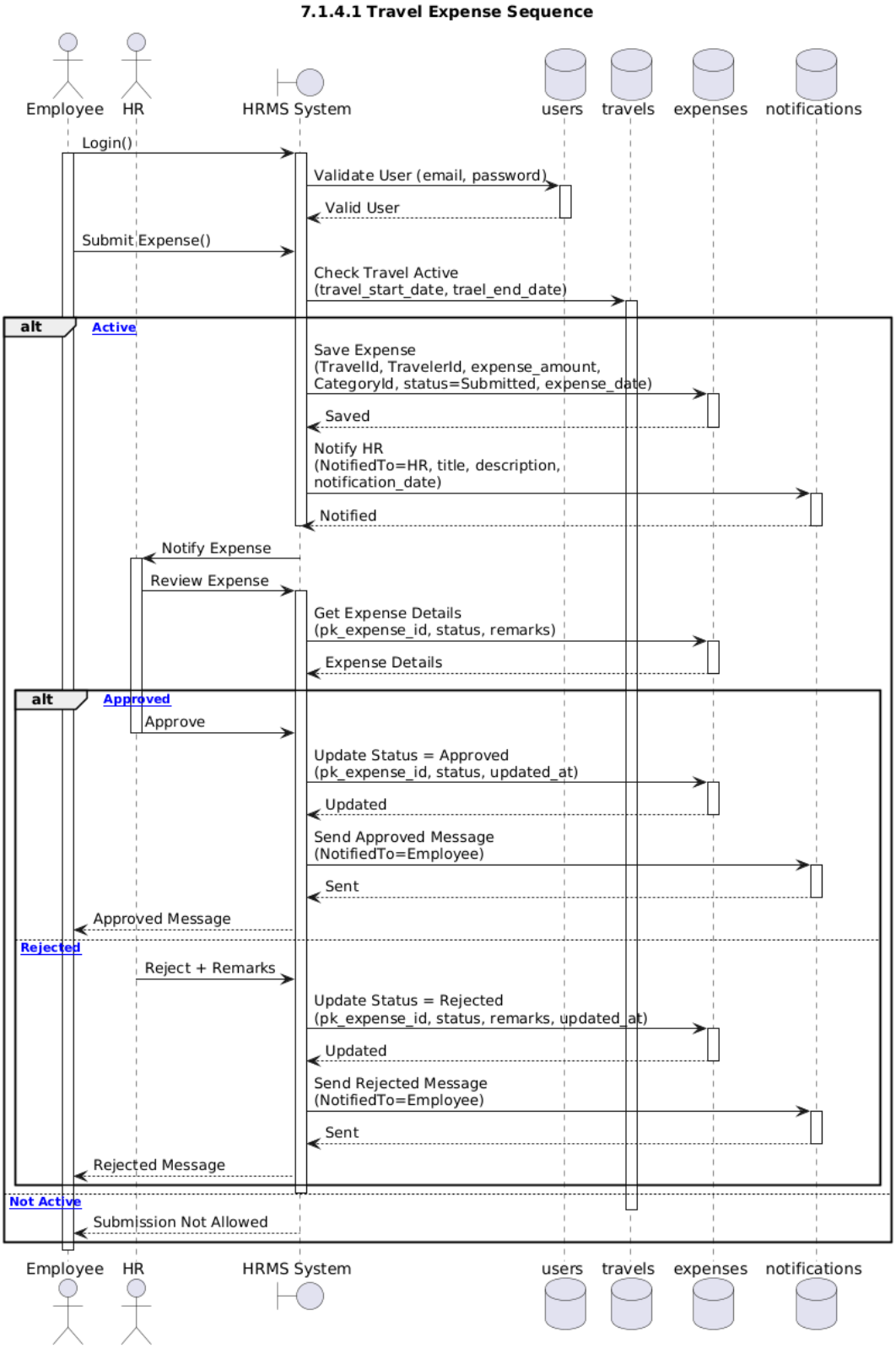


7.1.3 Class Diagram :

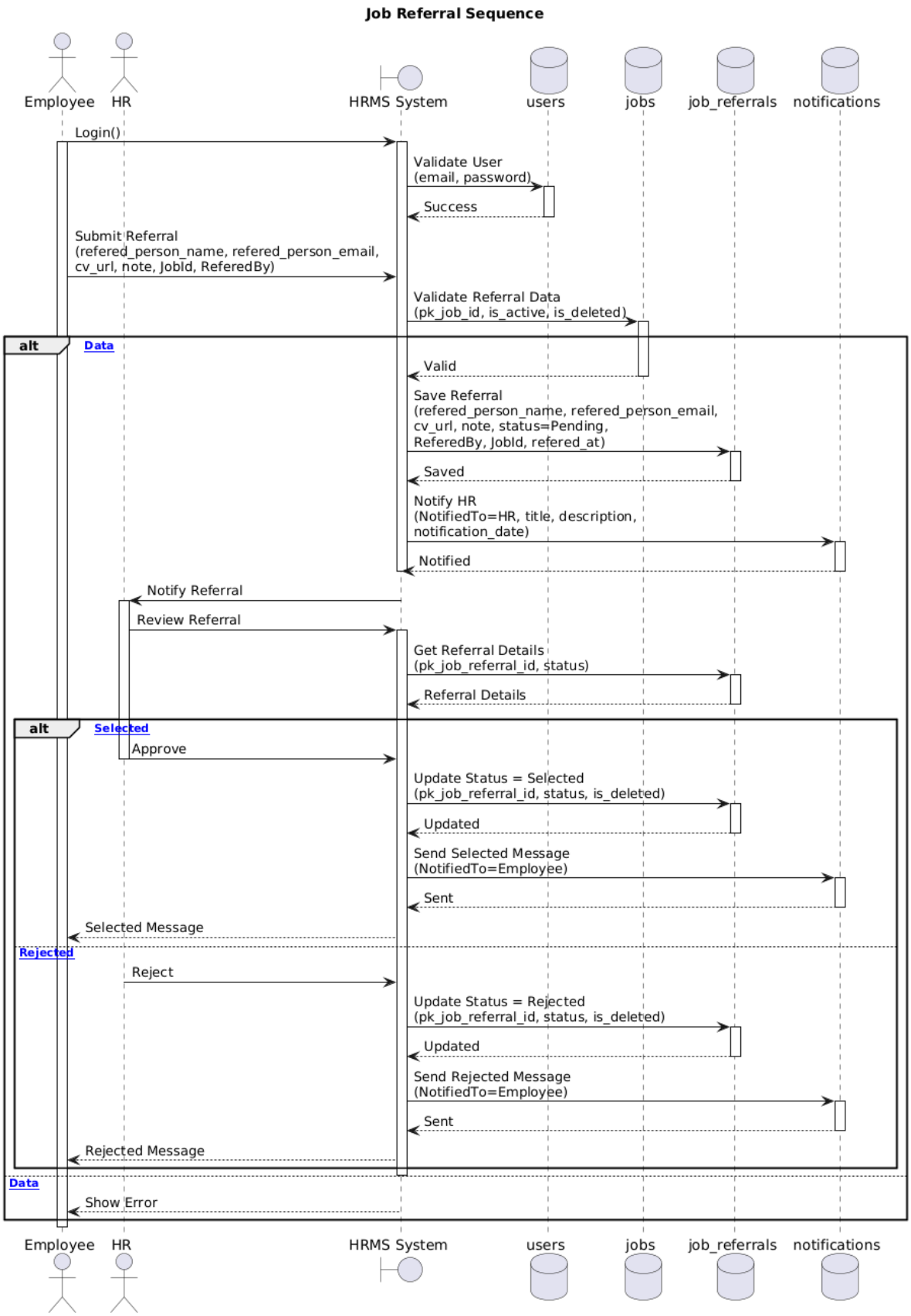


7.1.4 Sequence Diagram :

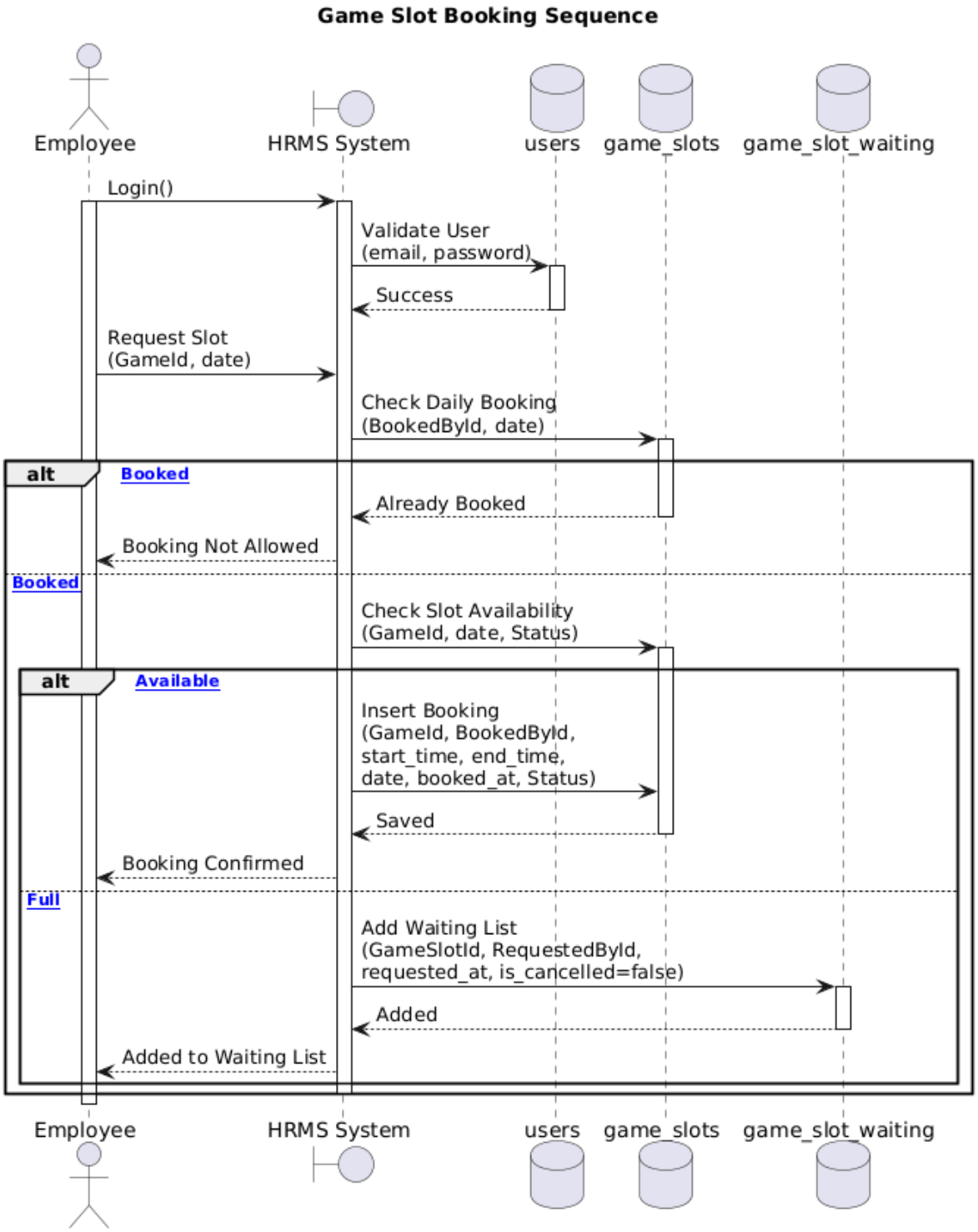
7.1.4.1 Travel Expense Sequence :



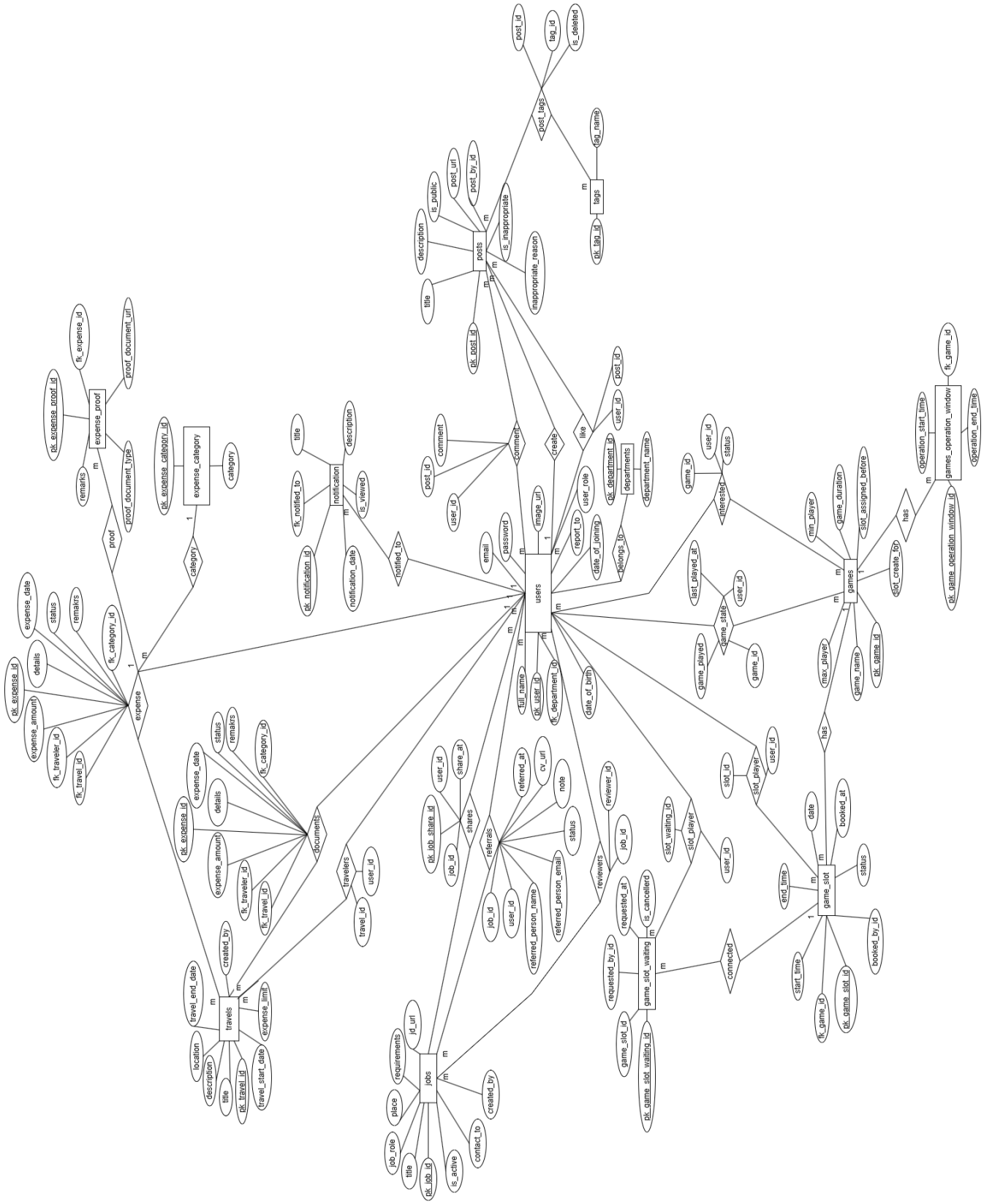
7.1.4.2 Job Referral Sequence :



7.1.4.3 Slot Booking Sequence :



7.2 ER Diagram :



7.3 Data Dictionary :

7.3.1 daily_celebrations : used to store daily celebration related data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_daily_celebration_id	INT	-	PK, Identity	Celebration ID
2	UserId	INT	-	FK(users)	Employee
3	celebration_type	INT	-	NOT NULL	Birthday / Anniversary
4	celebration_date	DATETIME2	-	NOT NULL	Celebration date

7.3.2 departments : used to store department of system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_department_id	INT	-	PK, Identity	Department ID
2	DepartmentName	NVARCHAR	30	NOT NULL	Department name
3	is_deleted	BIT	-	NOT NULL	Soft delete
4	created_at	DATETIME2	-	DEFAULT	Created time
5	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.3 expense_category : used to store expense category

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_category_id	INT	-	PK, Identity	Category ID
2	Category	NVARCHAR	50	NOT NULL	Expense category
3	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.4 **expense_proof :** used to store proof(document) related to expense

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_proof_id	INT	-	PK, Identity	Proof ID
2	ExpenseId	INT	-	FK(expenses)	Expense reference
3	proof_document_url	NVARCHAR	500	NOT NULL	Document URL
4	expense_document_type	NVARCHAR	50	NOT NULL	Document type
5	remarks	NVARCHAR	200	NOT NULL	Remarks

7.3.5 **expense :** used to store expense related data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_expense_id	INT	-	PK, Identity	Expense ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Employee
4	expense_amount	DECIMAL	10,2	NOT NULL	Amount
5	CategoryId	INT	-	FK(expense_category)	Category
6	status	NVARCHAR	20	NOT NULL	Draft/Submitted/ Approved/Rejected
7	remarks	NVARCHAR	100	NOT NULL	Remarks
8	details	NVARCHAR	200	NOT NULL	Details
9	expense_date	DATETIME2	-	NOT NULL	Expense date
10	is_deleted	BIT	-	NOT NULL	Soft delete
11	created_at	DATETIME2	-	DEFAULT	Created time
12	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.6 **game_operation_window :** used to store operationg hours of game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_operation_window_id	INT	-	PK, Identity	Window ID
2	GameId	INT	-	FK(games)	Game
3	operational_start_time	TIME	-	NOT NULL	Start time
4	operational_end_time	TIME	-	NOT NULL	End time
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	created_at	DATETIME2	-	NOT NULL	Created time
7	updated_at	DATETIME2	-	NOT NULL	Updated time

7.3.7 **game_slot_players :** used to store players of the slot of game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_player_id	INT	-	PK, Identity	Game slot player ID
2	SlotId	INT	-	FK(game_slots)	Game slot
3	PlayerId	INT	-	FK(users)	Player

7.3.8 **game_slot_waiting :** used to store waiting list of the game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_waiting_id	INT	-	PK, Identity	Waiting ID
2	GameSlotId	INT	-	FK(game_slots)	Game slot
3	RequestedById	INT	-	FK(users)	Requested by
4	requested_at	DATETIME2	-	DEFAULT	Requested time
5	is_cancelled	BIT	-	NOT NULL	Cancellation flag

7.3.9 **game_slot_waiting_players :** used to store player listing who are in waiting list of specific game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_waiting_player_id	INT	-	PK, Identity	Waiting player ID
2	GameSlotWaitingId	INT	-	FK(game_slot_waiting)	Waiting slot
3	PlayerId	INT	-	FK(users)	Player

7.3.10 **game_slot :** used to store auto-generated slot of the game.

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_slot_id	INT	-	PK, Identity	Slot ID
2	GameId	INT	-	FK(games)	Game
3	start_time	TIME	-	NOT NULL	Start time
4	end_time	TIME	-	NOT NULL	End time
5	date	DATETIME2	-	NOT NULL	Slot date
6	BookedById	INT	-	FK(users)	Booked by
7	booked_at	DATETIME2	-	NULL	Booked time
8	Status	INT	-	NOT NULL	Slot status

7.3.11 **games :** used to store game of the system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_game_id	INT	-	PK, Identity	Game ID
2	game_name	NVARCHAR	30	NOT NULL	Game name
3	max_player_per_game	INT	-	NOT NULL	Max players
4	min_player_per_game	INT	-	NOT NULL	Min players
5	duration_in_minutes	INT	-	NOT NULL	Duration
6	slot_assigned_before_x_minutes	INT	-	NOT NULL	Buffer time
7	slot_create_for_next_x_days	INT	-	NOT NULL	Slot generation days
8	is_deleted	BIT	-	NOT NULL	Soft delete
9	created_at	DATETIME2	-	DEFAULT	Created time
10	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.12 **job_referrals :** used to store job referrals on specific job

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_referral_id	INT	-	PK, Identity	Referral ID
2	refered_person_name	NVARCHAR	50	NOT NULL	Candidate name
3	refered_person_email	NVARCHAR	50	NOT NULL	Candidate email
4	cv_url	NVARCHAR	500	NOT NULL	CV URL
5	note	NVARCHAR	500	NOT NULL	Note
6	status	NVARCHAR	20	NOT NULL	Status
7	ReferedBy	INT	-	FK(users)	Referred by
8	JobId	INT	-	FK(jobs)	Job
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	refered_at	DATETIME2	-	NOT NULL	Referral date

7.3.13 **job_reviewers :** used to store reviewers of specific job

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_reviewer_id	INT	-	PK, Identity	Job reviewer ID
2	JobId	INT	-	FK(jobs)	Job
3	ReviewerId	INT	-	FK(users)	Reviewer

7.3.14 **job_shared :** used to store job share records

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_shared_id	INT	-	PK, Identity	Job share ID
2	shared_to	NVARCHAR	50	NOT NULL	Shared email
3	shared_at	INT	-	FK(users)	Shared by
4	JobId	INT	-	FK(jobs)	Job
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	SharedAt	DATETIME2	-	NOT NULL	Shared date

7.3.15 **jobs :** used to store jobs of the system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_job_id	INT	-	PK, Identity	Job ID
2	title	NVARCHAR	50	NOT NULL	Job title
3	job_role	NVARCHAR	20	NOT NULL	Role
4	place	NVARCHAR	30	NOT NULL	Location
5	requirements	NVARCHAR	300	NOT NULL	Requirements
6	jd_url	NVARCHAR	500	NOT NULL	JD URL
7	CreatedBy	INT	-	FK(users)	Created by
8	ContactTo	INT	-	FK(users)	HR contact
9	is_active	BIT	-	NOT NULL	Active
10	is_deleted	BIT	-	NOT NULL	Soft delete
11	created_at	DATETIME2	-	DEFAULT	Created time
12	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.16 **notifications :** used to store notification of system

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_notification_id	INT	-	PK, Identity	Notification ID
2	NotifiedTo	INT	-	FK(users)	User
3	title	NVARCHAR	50	NOT NULL	Title
4	description	NVARCHAR	500	NOT NULL	Description
5	is_viewed	BIT	-	NOT NULL	Viewed
6	notification_date	DATETIME2	-	NOT NULL	Date

7.3.17 **post_comments :** used to store comment on posts.

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_comment_id	INT	-	PK, Identity	Comment ID
2	comment	NVARCHAR	500	NOT NULL	Comment text
3	PostId	INT	-	FK(posts)	Post
4	CommentById	INT	-	FK(users)	Commented by
5	is_deleted	BIT	-	NOT NULL	Soft delete
6	created_at	DATETIME2	-	DEFAULT	Created time
7	updated_at	DATETIME2	-	DEFAULT	Updated time

7.3.18 **post_likes :** used to store like on the post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_like_id	INT	-	PK, Identity	Like ID
2	PostId	INT	-	FK(posts)	Post
3	LikedById	INT	-	FK(users)	Liked by
4	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.19 **post_tages :** used to store tag of the post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_tag_id	INT	-	PK, Identity	Post tag ID
2	PostId	INT	-	FK(posts)	Post
3	TagId	INT	-	FK(tags)	Tag
4	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.20 **posts :** used to store post created by users

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_post_id	INT	-	PK, Identity	Post ID
2	post_url	NVARCHAR	500	NOT NULL	Media URL
3	PostById	INT	-	FK(users)	Posted by
4	title	NVARCHAR	100	NOT NULL	Title
5	description	NVARCHAR	1000	NOT NULL	Description
6	is_public	BIT	-	NOT NULL	Visibility
7	is_inappropriate	BIT	-	NOT NULL	Moderation flag
8	inappropriate_reason	NVARCHAR	500	NULL	Reason
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	created_at	DATETIME2	-	DEFAULT	Created
11	updated_at	DATETIME2	-	DEFAULT	Updated

7.3.21 **tags :** used to store common tag of system for post

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_tag_id	INT	-	PK, Identity	Tag ID
2	tag_name	NVARCHAR	30	NOT NULL	Tag name
3	IsDeleted	BIT	-	NOT NULL	Soft delete

7.3.22 **travel_documents :** used to store document of travel

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_travel_document_id	INT	-	PK, Identity	Document ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Traveler
4	document_url	NVARCHAR	400	NOT NULL	Document URL
5	document_name	NVARCHAR	30	NOT NULL	Document name
6	document_type	NVARCHAR	50	NOT NULL	Document type
7	UploadedBy	INT	-	FK(users)	Uploaded by
8	uploaded_at	DATETIME2	-	NOT NULL	Upload time
9	is_deleted	BIT	-	NOT NULL	Soft delete

7.3.23 **travellers :** used to store travellers of specific travel

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_traveler_id	INT	-	PK, Identity	Traveler ID
2	TravelId	INT	-	FK(travels)	Travel
3	TravelerId	INT	-	FK(users)	Traveler
4	is_deletd	BIT	-	NOT NULL	Soft delete

7.3.24 **travels :** used to store travel data

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_travel_id	INT	-	PK, Identity	Travel ID
2	title	NVARCHAR	50	NOT NULL	Title
3	description	NVARCHAR	300	NOT NULL	Description
4	travel_start_date	DATETIME2	-	NOT NULL	Start date
5	trael_end_date	DATETIME2	-	NOT NULL	End date
6	location	NVARCHAR	50	NOT NULL	Location
7	CreatedBy	INT	-	FK(users)	Created by
8	expense_max_amount_limit	DECIMAL	10,2	NOT NULL	Limit
9	is_deleted	BIT	-	NOT NULL	Soft delete
10	created_at	DATETIME2	-	DEFAULT	Created
11	updated_at	DATETIME2	-	DEFAULT	Updated

7.3.25 **user_game_interest :** used to store game interest of user in specific game

Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_game_id	INT	-	PK, Identity	Interest ID
2	GameId	INT	-	FK(games)	Game
3	UserId	INT	-	FK(users)	User
4	status	NVARCHAR	20	NOT NULL	Interest status

7.3.26 **user_game_state :** used to store state of user on specific game

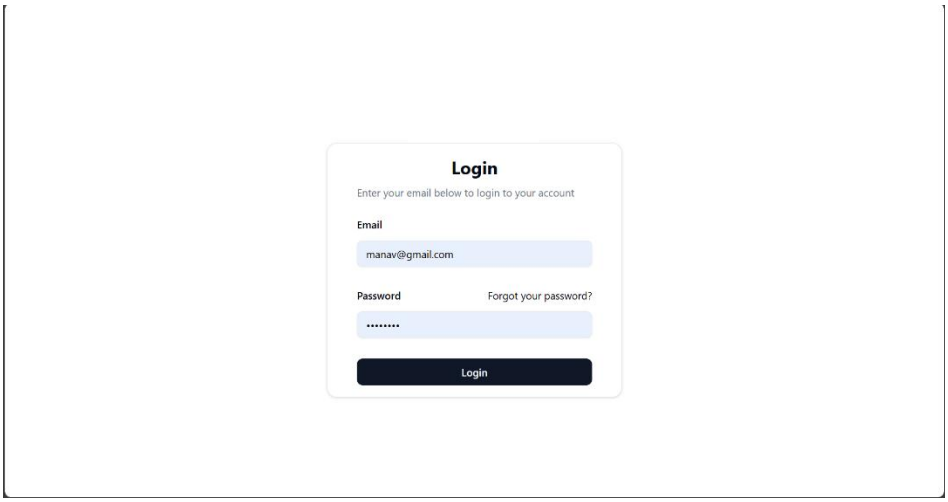
Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_game_state_id	INT	-	PK, Identity	Game state ID
2	GameId	INT	-	FK(games)	Game
3	UserId	INT	-	FK(users)	User
4	game_played	INT	-	NOT NULL	Times played
5	last_played_at	DATETIME2	-	NOT NULL	Last played time

7.3.27 **users :** used to store data related to system user

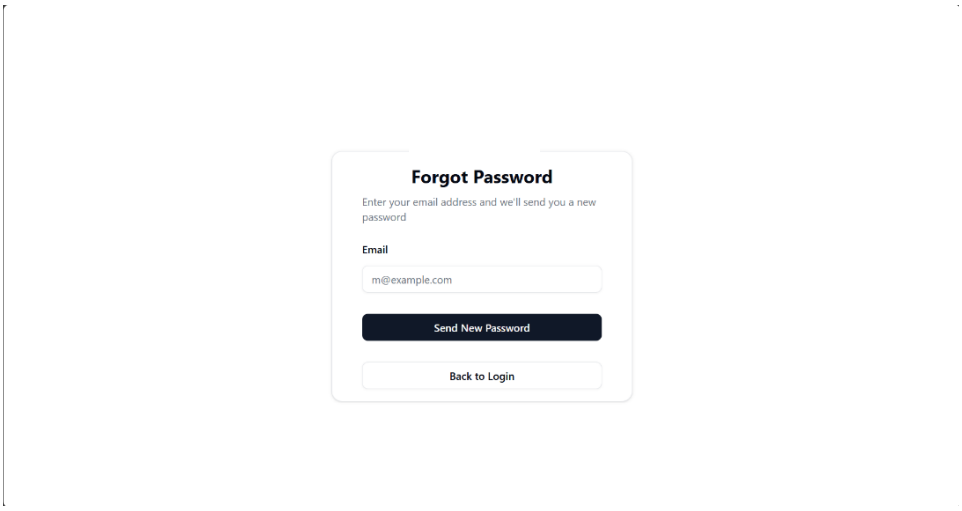
Sr.No	Column Name	Data Type	Size	Constraint	Description
1	pk_user_id	INT	-	PK, Identity	User ID
2	full_name	NVARCHAR	150	NOT NULL	Full name
3	email	NVARCHAR	50	UNIQUE, NOT NULL	Email
4	password	NVARCHAR	500	NOT NULL	Password
5	image_url	NVARCHAR	500	NOT NULL	Profile image
6	user_role	NVARCHAR	15	NOT NULL	Role
7	date_of_birth	DATETIME2	-	NOT NULL	DOB
8	date_of_joining	DATETIME2	-	NOT NULL	DOJ
9	ReportTo	INT	-	FK(users)	Manager
10	DepartmentId	INT	-	FK(departments)	Department
11	designation	NVARCHAR	30	NOT NULL	Designation
12	is_deleted	BIT	-	NOT NULL	Soft delete
13	created_at	DATETIME2	-	DEFAULT	Created
14	updated_at	DATETIME2	-	DEFAULT	Updated

8. IMPLEMENTATION OF SYSTEM

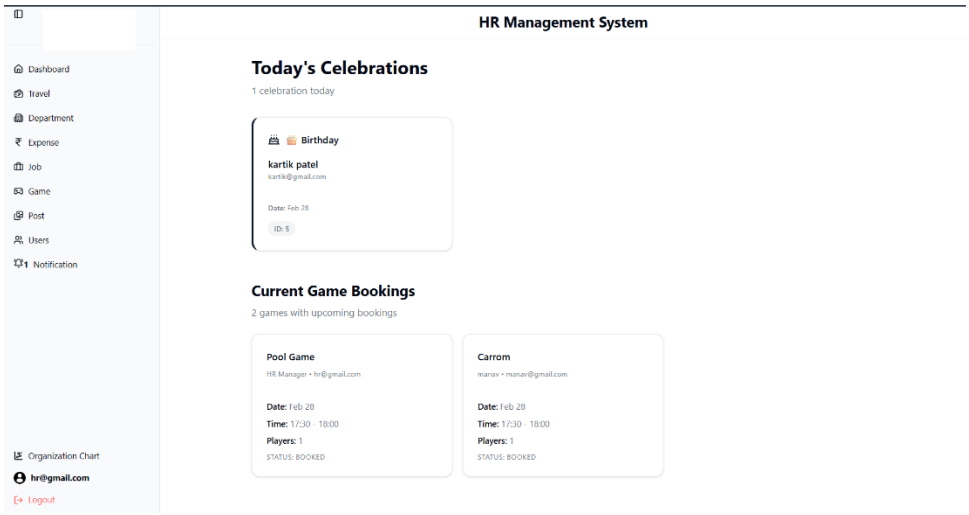
8.1 Screen Snapshots :



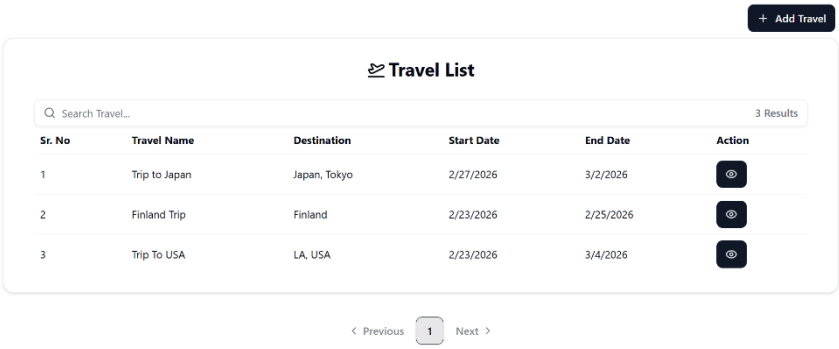
- Login Page Used to login into system



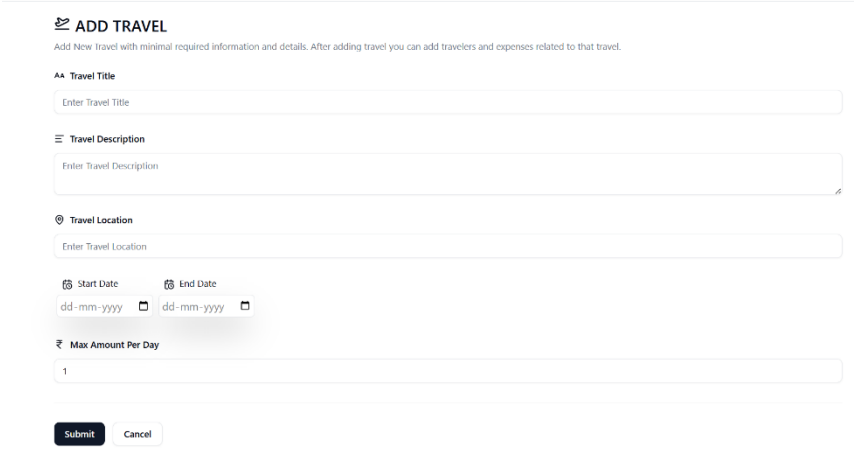
- Forget password to generate new password and get in email



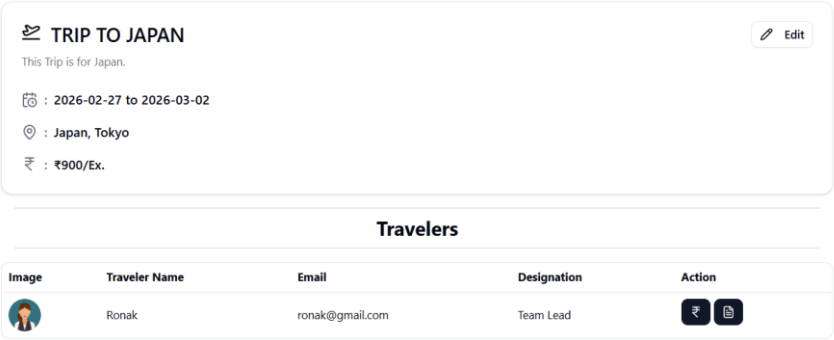
- Dashboard for HR of system



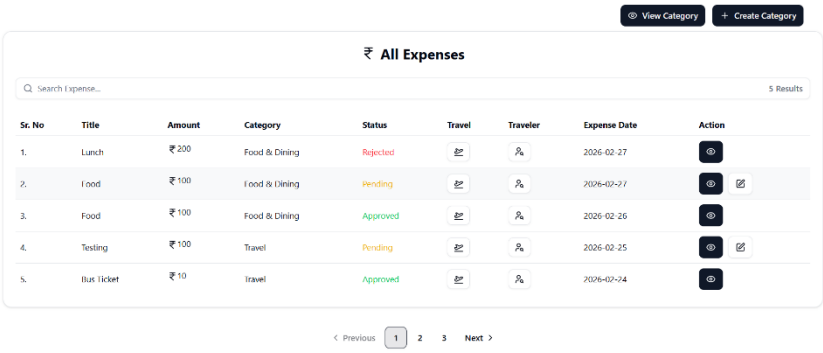
- List All Travel Created by Logged in HR



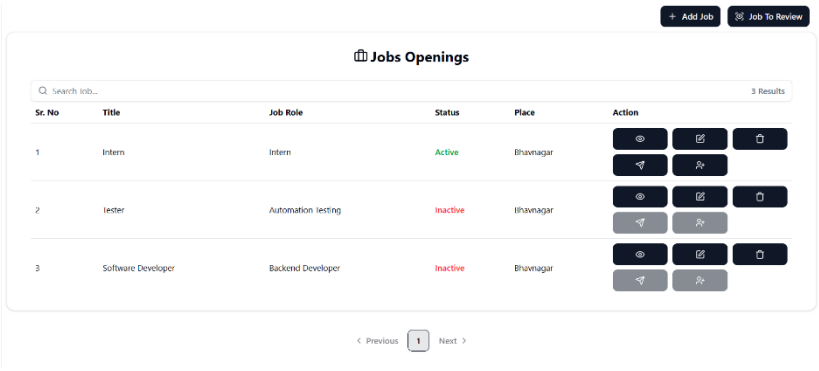
- Travel Create Page



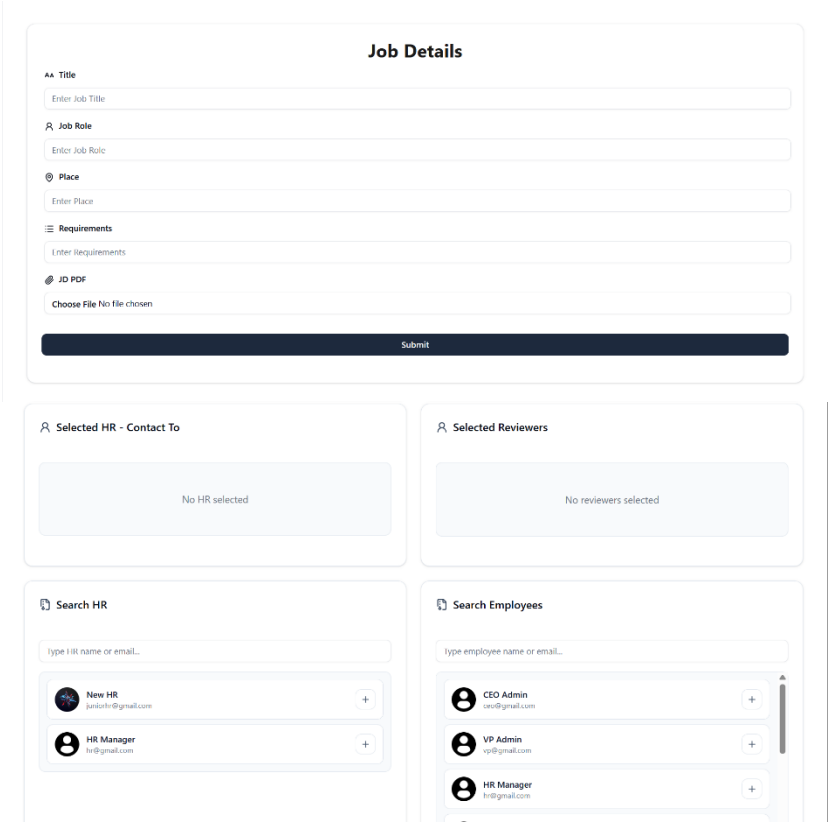
- Travel Detail Page with travellers



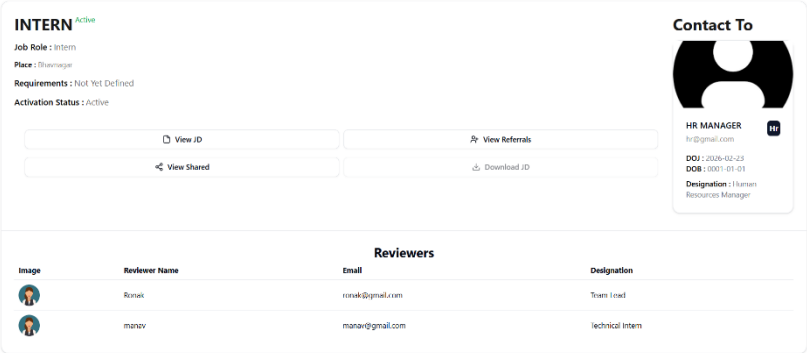
- All Travel Expenses of Traveller for changing status of it and preview proof of expense.



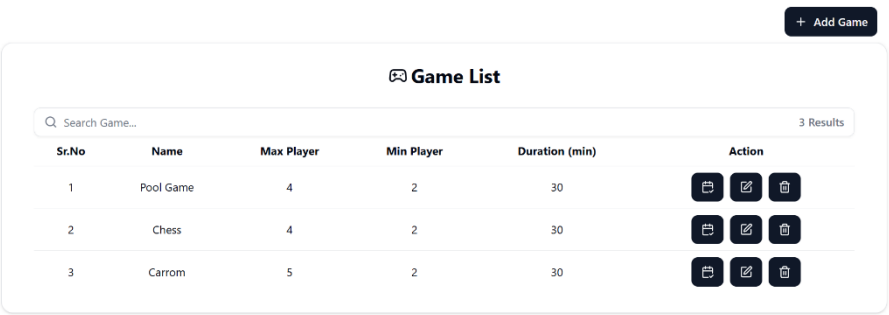
- Job Opening Created by logged in HR.



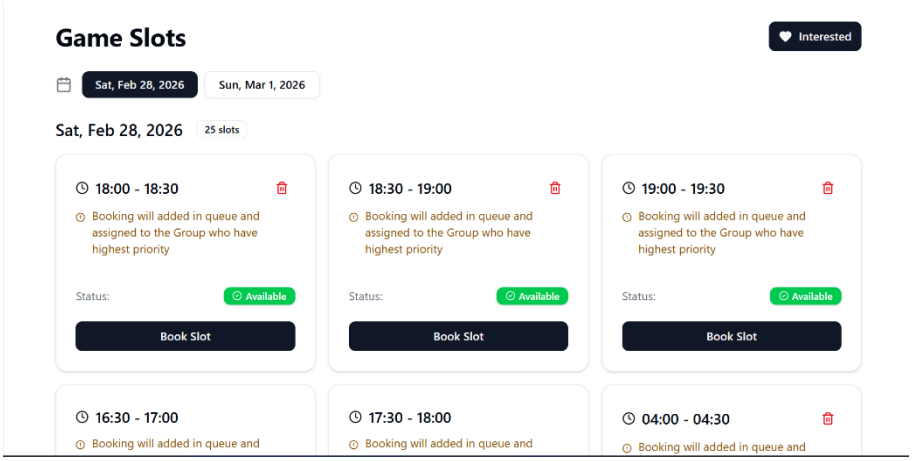
- Create Job For Opening.



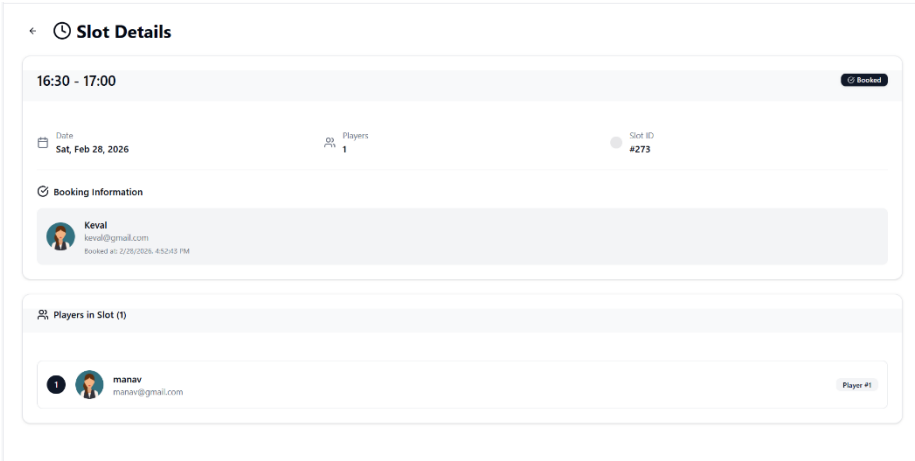
- Job Detail with reviewers and Contact to Detail



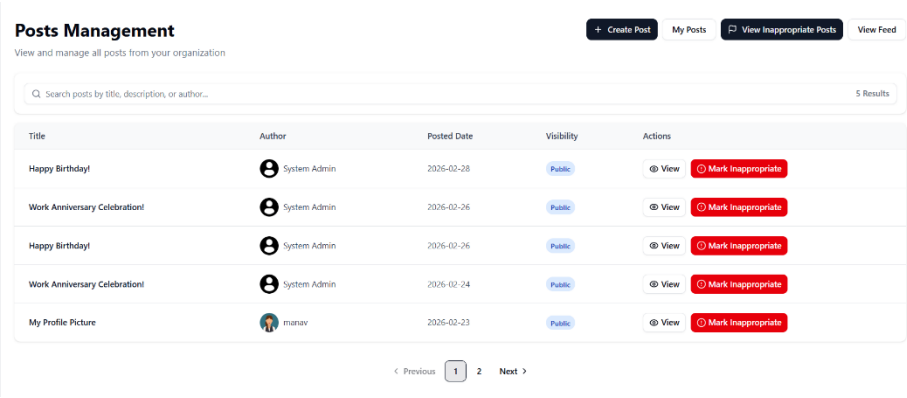
- All Game Of System.



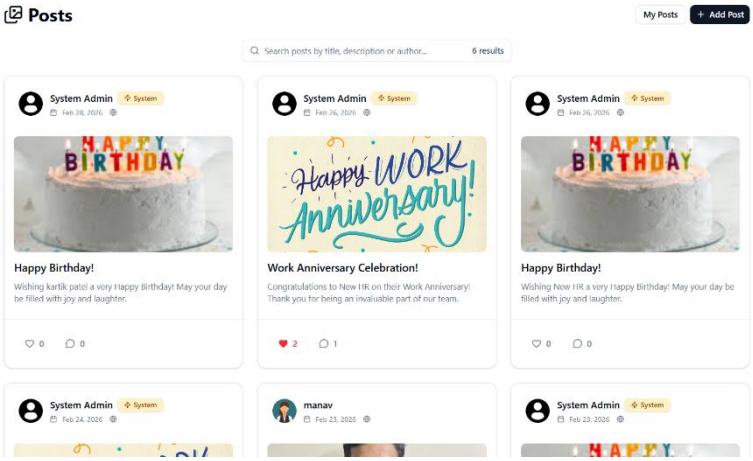
- Game Slot



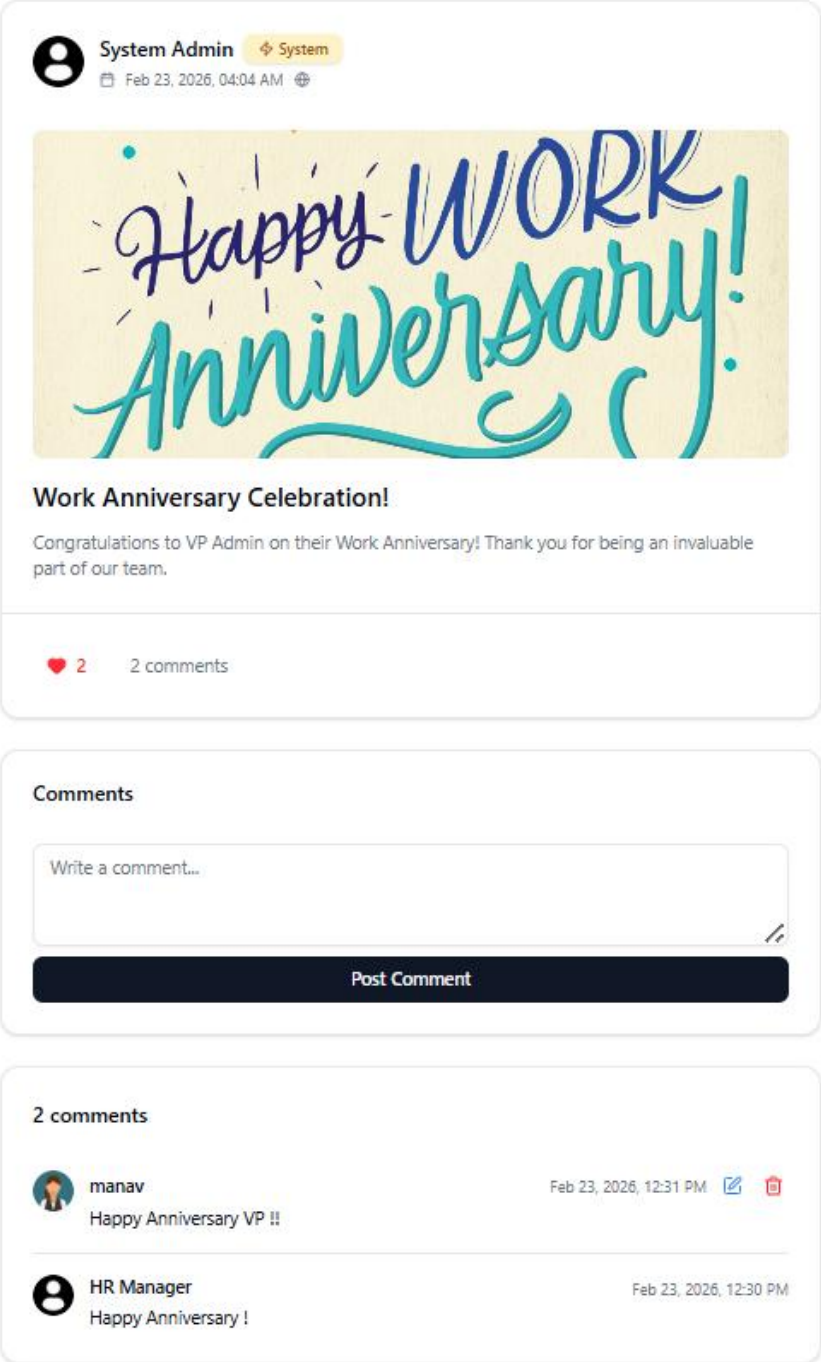
- Game Slot Booking Details with who booked and player of that slot.



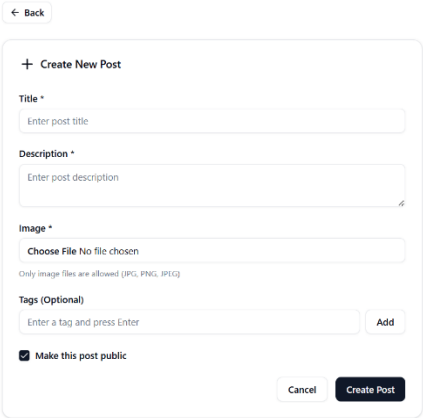
- Manage Post By HR to make post inappropriate or view details.



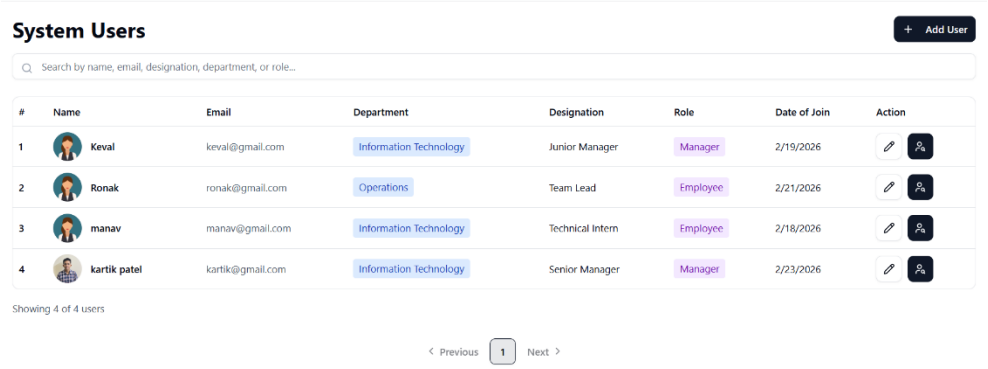
- Post Feed For All User who are logged in system



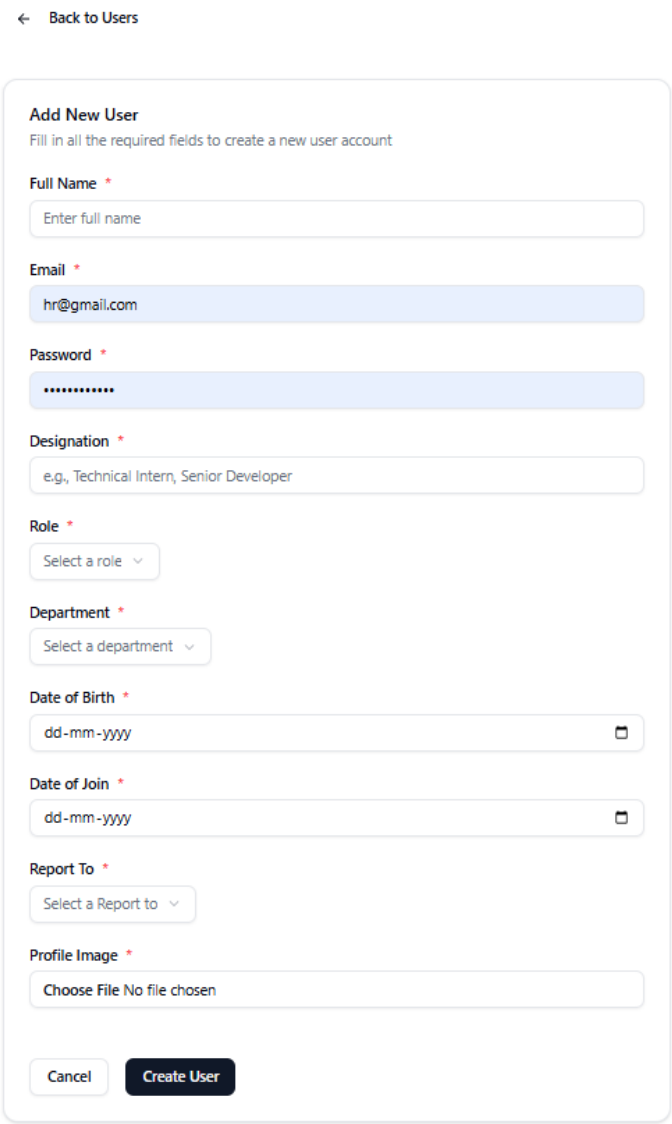
- Post Detail with Comment and like count.



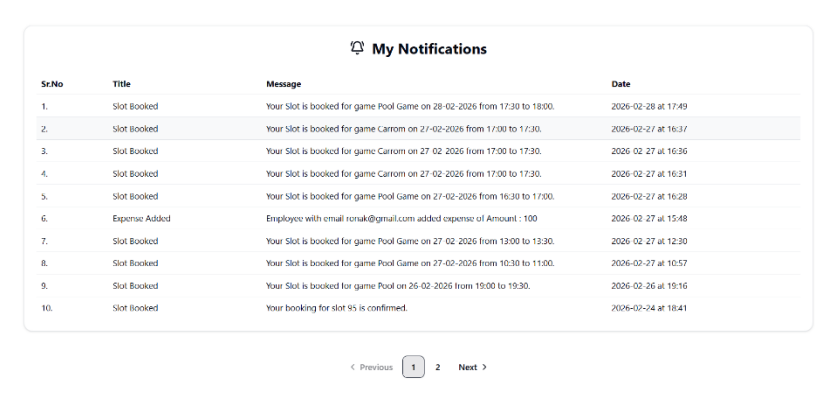
➢ Create Post Form.



- Manage User Page for HR



- Create User Page.

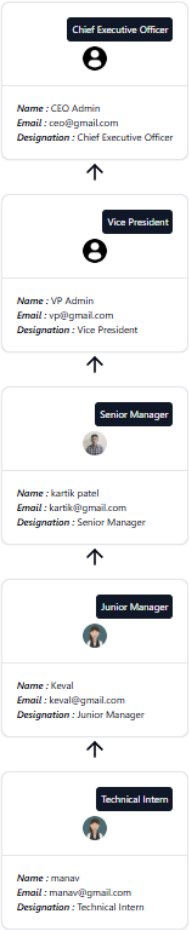


My Notifications

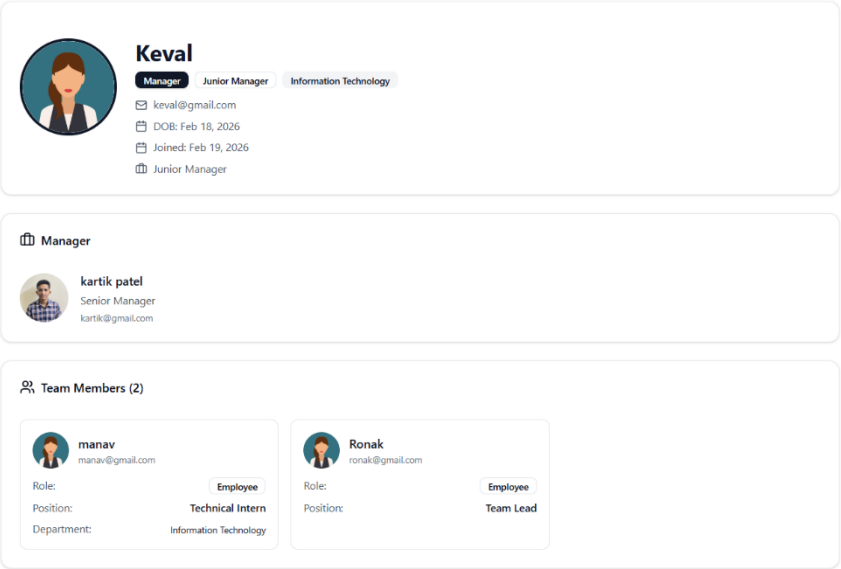
Sr.No	Title	Message	Date
1.	Slot Booked	Your Slot is booked for game Pool Game on 28-02-2026 from 17:30 to 18:00.	2026-02-28 at 17:49
2.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:37
3.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:36
4.	Slot Booked	Your Slot is booked for game Carrom on 27-02-2026 from 17:00 to 17:30.	2026-02-27 at 16:31
5.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 16:30 to 17:00.	2026-02-27 at 16:28
6.	Expense Added	Employee with email ronak@gmail.com added expense of Amount : 100	2026-02-27 at 15:48
7.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 13:00 to 13:30.	2026-02-27 at 12:30
8.	Slot Booked	Your Slot is booked for game Pool Game on 27-02-2026 from 10:30 to 11:00.	2026-02-27 at 10:57
9.	Slot Booked	Your Slot is booked for game Pool on 26-02-2026 from 19:00 to 19:30.	2026-02-26 at 19:16
10.	Slot Booked	Your booking for slot 85 is confirmed.	2026-02-24 at 18:41

Previous 1 2 Next

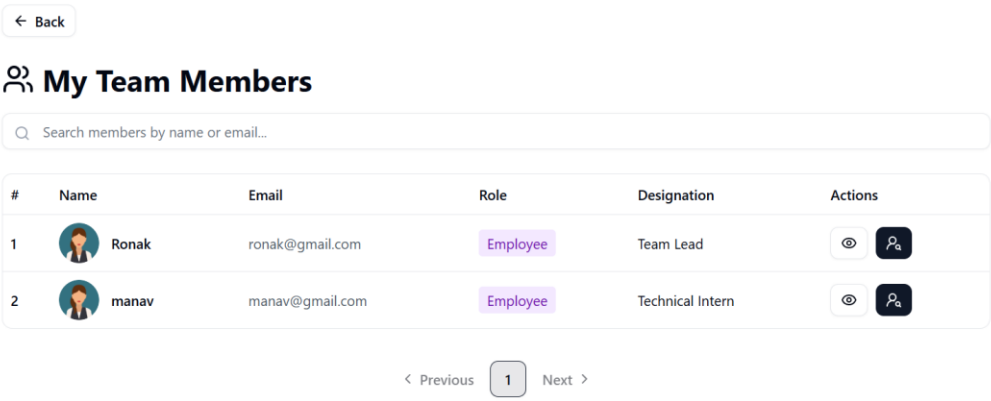
- Notification of current logged in user.



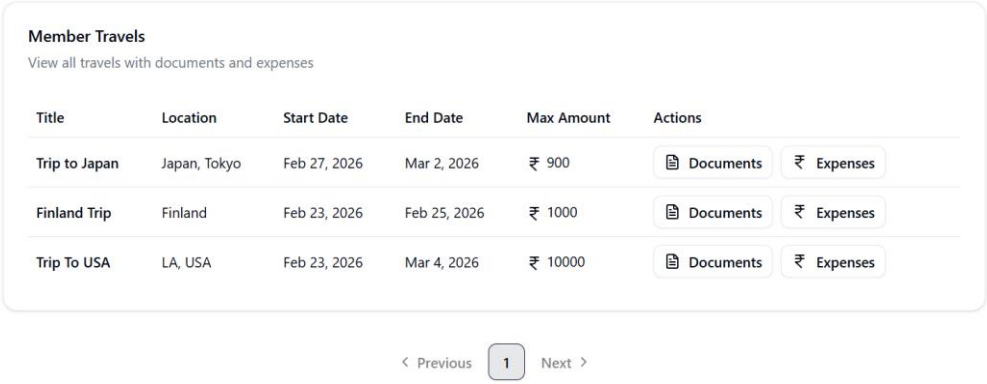
- Organization Chart



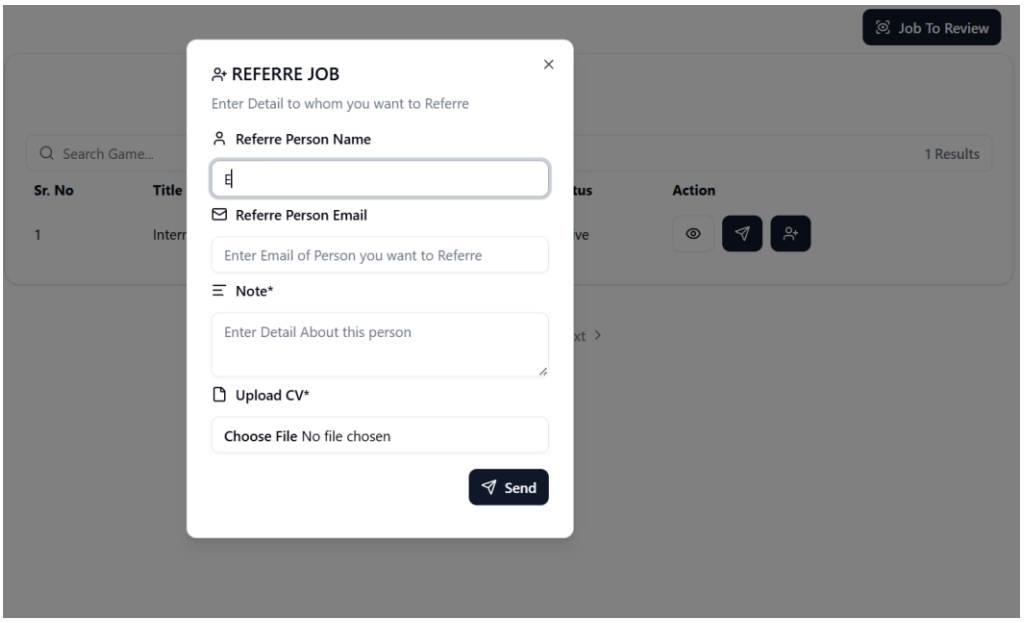
- User Profile with his Manager and His Team(if user is manager)



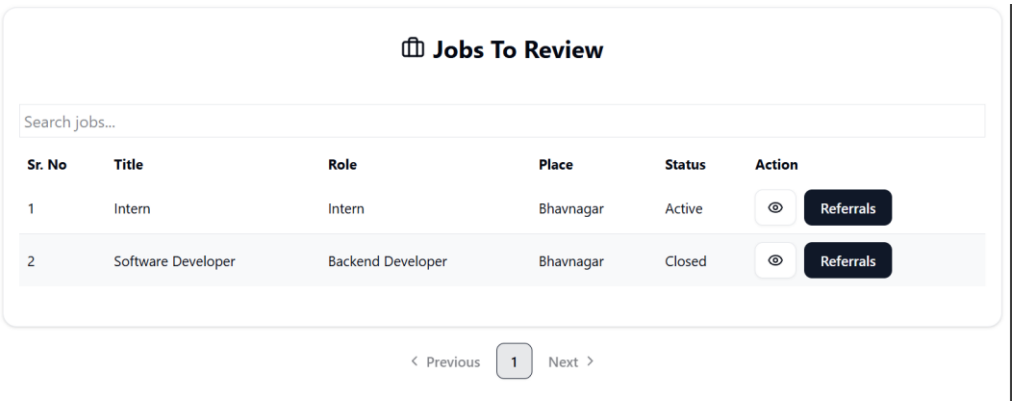
- My Team Detail for Manager



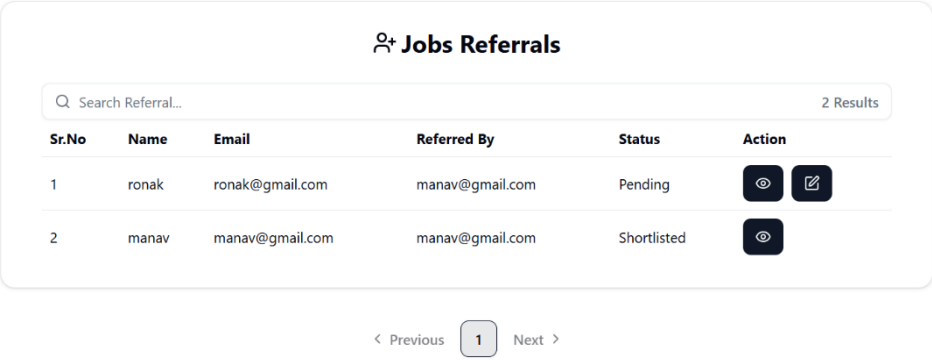
- Team Member Travel Detail



- Job Referral Create Form



- Job To review Page for Job Reviewers



- Job Referrals for Review.

8.2 Reports :

8.3 Testing :

8.3.1 Testing Strategy :

- **Black-Box Testing**

Black box testing is a testing technique in which the internal workings of the software are not known to the tester. The tester only focuses on the input and output of the software. Black box testing is mainly focused on testing the functionality of the software, ensuring that it meets the requirements and specifications. Black box testing uses methods like equivalence partitioning, boundary value analysis, and error guessing to create test cases.

8.3.2 Test cases and results :

1. **Add Travel Form:**

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add Travel Form	Travel Title	Valid Travel Title	Business Trip	Accepted	Accepted	Pass
2	Add Travel Form	Travel Title	Empty Travel Title		Validation Error	Validation Error	Pass
3	Add Travel Form	Travel Title	Title exceeds maximum length	300 characters	Validation Error	Validation Error	Pass
4	Add Travel Form	Travel Description	Valid Description	Client meeting in Delhi office	Accepted	Accepted	Pass
5	Add Travel Form	Travel Description	Empty Description		Validation Error	Validation Error	Pass
6	Add Travel Form	Travel Location	Valid Location	Delhi	Accepted	Accepted	Pass
7	Add Travel Form	Travel Location	Empty Location		Validation Error	Validation Error	Pass
8	Add Travel Form	Start Date	Valid Start Date	10-06-2026	Accepted	Accepted	Pass
9	Add Travel Form	Start Date	Empty Start Date		Validation Error	Validation Error	Pass
10	Add Travel Form	End Date	Valid End Date	15-06-2026	Accepted	Accepted	Pass
11	Add Travel Form	End Date	End Date earlier than Start Date	Start: 15-06-2026 End: 10-06-2026	Validation Error	Validation Error	Pass
12	Add Travel Form	End Date	Empty End Date		Validation Error	Validation Error	Pass
13	Add Travel Form	Max Amount Per Day	Valid Amount	2000	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
14	Add Travel Form	Max Amount Per Day	Empty Amount		Validation Error	Validation Error	Pass
15	Add Travel Form	Max Amount Per Day	Negative Amount	-500	Validation Error	Validation Error	Pass
16	Add Travel Form	Max Amount Per Day	Non-numeric input	abc	Validation Error	Validation Error	Pass
17	Add Travel Form	Submit Button	Submit with valid data	Valid Form Data	Travel record created successfully	Travel record created successfully	Pass
18	Add Travel Form	Cancel Button	Click cancel button	—	User redirected to travel list page	Redirect successful	Pass

2. Job Opening Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Job Details Form	Title	Valid Job Title	Software Engineer	Accepted	Accepted	Pass
2	Job Details Form	Title	Empty Job Title		Validation Error	Validation Error	Pass
3	Job Details Form	Title	Very long title	300 characters	Validation Error	Validation Error	Pass
4	Job Details Form	Job Role	Valid Job Role	Backend Developer	Accepted	Accepted	Pass
5	Job Details Form	Job Role	Empty Job Role		Validation Error	Validation Error	Pass
6	Job Details Form	Place	Valid Place	Ahmedabad	Accepted	Accepted	Pass
7	Job Details Form	Place	Empty Place		Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
8	Job Details Form	Requirements	Valid Requirements	Java, Spring Boot, SQL	Accepted	Accepted	Pass
9	Job Details Form	Requirements	Empty Requirements		Validation Error	Validation Error	Pass
10	Job Details Form	JD PDF	Upload valid PDF file	JD.pdf	Accepted	Accepted	Pass
11	Job Details Form	JD PDF	Upload invalid file type	JD.png	Validation Error	Validation Error	Pass
12	Job Details Form	JD PDF	No file uploaded		Validation Error	Validation Error	Pass
13	HR Selection Panel	Search HR	Search HR by name	Komal Patel	HR appears in list	HR appears in list	Pass
14	HR Selection Panel	Search HR	Search HR by email	hr@gmail.com	HR appears in list	HR appears in list	Pass
15	HR Selection Panel	Select HR	Add HR using + button	Komal Patel	HR added to "Selected HR" section	HR added successfully	Pass
16	HR Selection Panel	Select HR	Select multiple HR users	Komal Patel, Falguni Patel	Both HR users added	Added successfully	Pass
17	HR Selection Panel	Select HR	No HR selected		Validation Error	Validation Error	Pass
18	Reviewer Selection Panel	Search Employees	Search employee by name	CEO	Employee displayed in list	Employee displayed	Pass
19	Reviewer Selection Panel	Search Employees	Search employee by email	vp@gmail.com	Employee displayed	Employee displayed	Pass
20	Reviewer Selection Panel	Select Reviewer	Add reviewer using + button	VP	Reviewer added to selected list	Reviewer added	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
21	Reviewer Selection Panel	Select Reviewer	Add multiple reviewers	VP, System Admin	Both reviewers added	Added successfully	Pass
22	Reviewer Selection Panel	Select Reviewer	No reviewer selected		Validation Error	Validation Error	Pass
23	Job Details Form	Submit Button	Submit with valid inputs	Valid job details	Job created successfully	Job created successfully	Pass
24	Job Details Form	Submit Button	Submit with missing fields	Missing title	Validation error displayed	Validation error displayed	Pass

3. Game Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Create New Game	Name	Valid Game Name	Chess	Accepted	Accepted	Pass
2	Create New Game	Name	Empty Game Name		Validation Error	Validation Error	Pass
3	Create New Game	Name	Game name too long	300 characters	Validation Error	Validation Error	Pass
4	Create New Game	Max Player	Valid Max Player	4	Accepted	Accepted	Pass
5	Create New Game	Max Player	Empty Max Player		Validation Error	Validation Error	Pass
6	Create New Game	Max Player	Negative value	-2	Validation Error	Validation Error	Pass
7	Create New Game	Max Player	Non-numeric value	abc	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
8	Create New Game	Min Player	Valid Min Player	2	Accepted	Accepted	Pass
9	Create New Game	Min Player	Empty Min Player		Validation Error	Validation Error	Pass
10	Create New Game	Min Player	Min Player greater than Max Player	Min=5 Max=4	Validation Error	Validation Error	Pass
11	Create New Game	Duration (Minutes)	Valid Duration	30	Accepted	Accepted	Pass
12	Create New Game	Duration (Minutes)	Empty Duration		Validation Error	Validation Error	Pass
13	Create New Game	Duration (Minutes)	Negative Duration	-10	Validation Error	Validation Error	Pass
14	Create New Game	Slot Assigned (Min)	Valid Slot Assigned	60	Accepted	Accepted	Pass
15	Create New Game	Slot Assigned (Min)	Empty Slot Assigned		Validation Error	Validation Error	Pass
16	Create New Game	Slot Assigned (Min)	Non-numeric value	abc	Validation Error	Validation Error	Pass
17	Create New Game	Slot Create For Next (Days)	Valid days value	7	Accepted	Accepted	Pass
18	Create New Game	Slot Create For Next (Days)	Empty days value		Validation Error	Validation Error	Pass
19	Create New Game	Slot Create For Next (Days)	Negative days value	-5	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
20	Create New Game	Create Slot Button	Submit valid game configuration	Valid form data	Game created and slots generated successfully	Game created successfully	Pass

4. Post Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Create New Post	Title	Valid Title	Team Achievement	Accepted	Accepted	Pass
2	Create New Post	Title	Empty Title		Validation Error	Validation Error	Pass
3	Create New Post	Title	Very long title	300 characters	Validation Error	Validation Error	Pass
4	Create New Post	Description	Valid Description	Our team successfully completed the project	Accepted	Accepted	Pass
5	Create New Post	Description	Empty Description		Validation Error	Validation Error	Pass
6	Create New Post	Description	Very long description	1000+ characters	Accepted	Accepted	Pass
7	Create New Post	Image	Valid image upload	post.jpg	Accepted	Accepted	Pass
8	Create New Post	Image	Upload PNG image	post.png	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
9	Create New Post	Image	Upload invalid file type	post.pdf	Validation Error	Validation Error	Pass
10	Create New Post	Image	No image uploaded		Validation Error	Validation Error	Pass
11	Create New Post	Tags	Add valid tag	celebration	Tag added successfully	Tag added	Pass
12	Create New Post	Tags	Add multiple tags	celebration, success	Tags added successfully	Tags added	Pass
13	Create New Post	Tags	Empty tag input		Tag not added	Tag not added	Pass
14	Create New Post	Public Checkbox	Checkbox checked	True	Post will be public	Post marked public	Pass
15	Create New Post	Public Checkbox	Checkbox unchecked	False	Post will be private	Post marked private	Pass
16	Create New Post	Create Post Button	Submit valid post	Valid form data	Post created successfully	Post created successfully	Pass
17	Create New Post	Create Post Button	Submit with missing fields	Missing title	Validation error displayed	Validation error displayed	Pass
18	Create New Post	Cancel Button	Click cancel button	—	Form closed without saving	Form closed	Pass

5. User Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add New User	Full Name	Valid full name	Kartik Patel	Accepted	Accepted	Pass
2	Add New User	Full Name	Empty full name		Validation Error	Validation Error	Pass
3	Add New User	Full Name	Name with numbers	Kartik123	Validation Error	Validation Error	Pass
4	Add New User	Email	Valid email	user@gmail.com	Accepted	Accepted	Pass
5	Add New User	Email	Empty email		Validation Error	Validation Error	Pass
6	Add New User	Email	Invalid email format	usergmail.com	Validation Error	Validation Error	Pass
7	Add New User	Email	Existing email	hr@gmail.com	Server Side Error	Server Side Error	Pass
8	Add New User	Password	Valid password	User@123	Accepted	Accepted	Pass
9	Add New User	Password	Empty password		Validation Error	Validation Error	Pass
10	Add New User	Password	Weak password	12345	Validation Error	Validation Error	Pass
11	Add New User	Designation	Valid designation	Software Engineer	Accepted	Accepted	Pass
12	Add New User	Designation	Empty designation		Validation Error	Validation Error	Pass
13	Add New User	Role	Select valid role	Employee	Accepted	Accepted	Pass
14	Add New User	Role	No role selected		Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
15	Add New User	Department	Select valid department	IT	Accepted	Accepted	Pass
16	Add New User	Department	No department selected		Validation Error	Validation Error	Pass
17	Add New User	Date of Birth	Valid date	10-05-2000	Accepted	Accepted	Pass
18	Add New User	Date of Birth	Empty DOB		Validation Error	Validation Error	Pass
19	Add New User	Date of Join	Valid joining date	01-07-2024	Accepted	Accepted	Pass
20	Add New User	Date of Join	Joining date before DOB	DOB: 10-05-2000 Join: 01-01-1995	Validation Error	Validation Error	Pass
21	Add New User	Report To	Select manager	VP	Accepted	Accepted	Pass
22	Add New User	Profile Image	Upload valid image	profile.jpg	Accepted	Accepted	Pass
23	Add New User	Profile Image	Upload invalid file	profile.pdf	Validation Error	Validation Error	Pass
24	Add New User	Profile Image	No image uploaded		Validation Error	Validation Error	Pass
25	Add New User	Create User Button	Submit valid form	Valid form data	User created successfully	User created successfully	Pass
26	Add New User	Cancel Button	Click cancel button	—	Form closes without saving	Form closed	Pass

6. Expense Create Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add New Expense	Details	Valid expense details	Taxi fare to airport	Accepted	Accepted	Pass
2	Add New Expense	Details	Empty expense details		Validation Error	Validation Error	Pass
3	Add New Expense	Details	Very long description	500+ characters	Accepted	Accepted	Pass
4	Add New Expense	Amount	Valid amount	1500	Accepted	Accepted	Pass
5	Add New Expense	Amount	Empty amount		Validation Error	Validation Error	Pass
6	Add New Expense	Amount	Negative amount	-100	Validation Error	Validation Error	Pass
7	Add New Expense	Amount	Non-numeric amount	abc	Validation Error	Validation Error	Pass
8	Add New Expense	Expense Date	Valid expense date	10-06-2026	Accepted	Accepted	Pass
9	Add New Expense	Expense Date	Empty expense date		Validation Error	Validation Error	Pass
10	Add New Expense	Expense Date	Date outside travel period	01-01-2030	Validation Error	Validation Error	Pass
11	Add New Expense	Category	Valid category selected	Travel	Accepted	Accepted	Pass
12	Add New Expense	Category	No category selected		Validation Error	Validation Error	Pass
13	Add New Expense	Proof	Upload valid PDF receipt	receipt.pdf	Accepted	Accepted	Pass
14	Add New Expense	Proof	Upload valid image receipt	receipt.jpg	Accepted	Accepted	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
15	Add New Expense	Proof	Upload unsupported file type	receipt.exe	Validation Error	Validation Error	Pass
16	Add New Expense	Proof	No file uploaded		Validation Error	Validation Error	Pass
17	Add New Expense	Submit Button	Submit with valid form data	Valid expense data	Expense submitted successfully	Expense submitted successfully	Pass
18	Add New Expense	Submit Button	Submit with missing required fields	Missing amount	Validation error displayed	Validation error displayed	Pass

7. Travel Document Add Form:

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
1	Add Travel Document	Document Name	Valid document name	Flight Ticket	Accepted	Accepted	Pass
2	Add Travel Document	Document Name	Empty document name		Validation Error	Validation Error	Pass
3	Add Travel Document	Document Name	Very long document name	300 characters	Validation Error	Validation Error	Pass
4	Add Travel Document	Document Type	Valid document type	Ticket	Accepted	Accepted	Pass
5	Add Travel Document	Document Type	Empty document type		Validation Error	Validation Error	Pass
6	Add Travel Document	Document Type	Invalid characters in	@@@	Validation Error	Validation Error	Pass

Test Case ID	Form/Screen	Field Name	Test Scenario	Input Data	Expected Result	Actual Result	Status (Pass / Fail)
			document type				
7	Add Travel Document	Document File	Upload valid PDF file	ticket.pdf	Accepted	Accepted	Pass
8	Add Travel Document	Document File	Upload image file	ticket.jpg	Accepted	Accepted	Pass
9	Add Travel Document	Document File	Upload unsupported file type	ticket.exe	Validation Error	Validation Error	Pass
10	Add Travel Document	Document File	No file uploaded		Validation Error	Validation Error	Pass
11	Add Travel Document	Upload Document Button	Submit valid document details	Valid document data	Document uploaded successfully	Document uploaded successfully	Pass
12	Add Travel Document	Upload Document Button	Submit with missing required fields	Missing document name	Validation error displayed	Validation error displayed	Pass
13	Add Travel Document	Cancel Button	Click cancel button	—	Modal closed without saving	Modal closed	Pass

9 . FUTURE ENHANCEMENT

- Mobile Application Support
- Multi-Level Approval Workflow
- AI-Based Resume Screening
- Real-Time Notifications System

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